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Achieving sustainable cultivation of coffee

Breeding and quality traits

Edited by Dr Philippe Lashermes
Institut de Recherche pour le Développement (IRD), France

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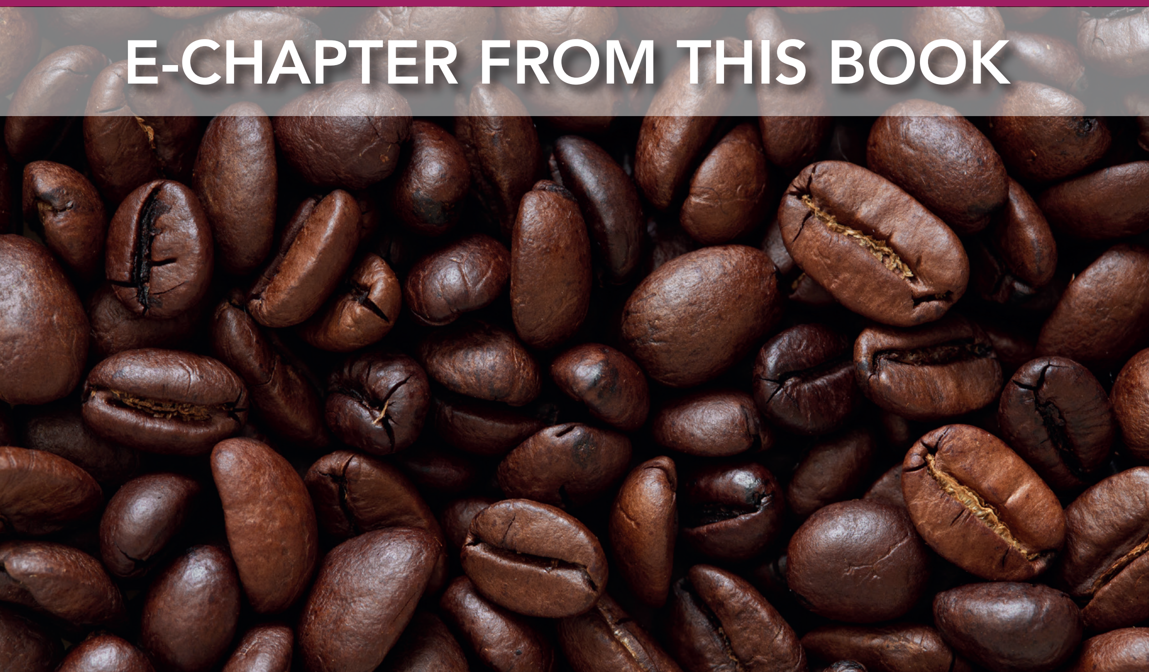

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Nutritional and health effects of coffee

Adriana Farah, Federal University of Rio de Janeiro, Brazil

1 Introduction

Good health and well-being are essential for all humans and depend upon good nutrition. Only when an individual has good health can he or she fully utilize their physical and mental potentials.

Since the beginning of humanity, plant foods have been used to promote health and prevent disease. Coffee has been exalted by people of different nations and times not only because of its distinctive aroma and taste but also due to its stimulating and health-promoting effects (Bizzo et al., 2015).

The earliest potential references to coffee consumption are found in the Old Testament, where a bean was referred to as a 'parched pulse', and the first written mention of coffee is attributed to Razes, a tenth-century Arabian physician, who indicated that coffee cultivation may have begun as early as 575 AD (Smith, 1987; Folmer et al., 2017). However, the first written documentation of the medicinal properties and uses of coffee was reported by the Middle Eastern physician, Avicenna (980–1037 AD), who used it as a decongestant, muscle relaxant and diuretic infusion. It is said that in the thirteenth century, a doctor-priest from Mocha, Sheikh Omar, also discovered coffee in Arabia and used it as a cure for many different types of illnesses (Ukers, 1935). The earliest coffee houses opened in Mecca in the fifteenth century, but were primarily reserved for religious purposes. After they were popularized, following a trip to Aleppo, Dr. Leonard Rauwolf, a German physician, introduced the beverage to Western Europe in the sixteenth century and referred to it as being 'almost as black as ink and very good in illness, chiefly that of the stomach' (Ukers, 1935; Folmer et al., 2017).

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Neither the Greek physician Hippocrates (460–377AD) nor the Roman physician Aelius Galenus (129–199 AD), who later expanded Galenic theories, ever mentioned coffee, as it was not consumed in their region at that time. However, the followers of Hippocratic–Galenic medicine, which dominated physiology from the fourth century BC to the nineteenth century, used it to balance the body's 'humours' in accordance with individual temperaments. Coffee was considered to be beneficial to people with a lymphatic or bilious temperament, whereas sanguine or nervous subjects were advised to use it more reservedly. Consequently, coffee houses, which were first introduced in Europe in the seventeenth century, were often recommended for those suffering from maladies as part of their treatment (Bizzo et al., 2015). In the eighteenth century, they also became places for social gathering and commerce, and over time, more coffee houses opened up and became popular (Folmer et al., 2017).

Despite its recognition as a medicinal agent or simply a beverage with an attractive taste, throughout history, coffee has occasionally been seen as a villain, and to date, remainders of this reputation still exist. The perception of coffee as an intoxicating drug and the sensitivity of some people to caffeine are the main reasons for this, along with past discussions on its potential contribution to the development of cancer or other diseases. However, over the last few decades, the appearance of modern scientific technology, in combination with large and reliable databases and sophisticated statistics, has enabled the separation of confounding factors in epidemiological studies such as existing medical conditions, smoking or a poor-quality diet. Additionally, an increasing number of studies have proved that despite its nutritional limitations, coffee is a complex mixture of bioactive substances that may act together to help prevent diseases when consumed appropriately. In view of this, our understanding of coffee and its healthful properties has changed dramatically. Currently, the general opinion is that moderate coffee consumption is not associated with increased long-term risks amongst healthy individuals and can be incorporated into a healthy and diverse diet, in combination with other healthful behaviours (US Department of Agriculture – USDA, 2015).

This chapter presents a brief literature review of the nutritional and main health-related aspects of regular coffee consumption.

2 Nutrients and bioactive compounds of coffee

The chemical composition of roasted coffee beans has been detailed in previous chapters. In summary, they contain approximately 43% carbohydrates (of which 70–85% is comprised of polysaccharides, arabinogalactans, mannans and glucan, and the remaining amount includes sucrose, reducing sugars, lignins and pectins); 7.5–10% proteins; other nitrogenous compounds (1% caffeine, 0.7–1% trigonelline and 0.01–0.04% nicotinic acid); 10–15% lipids (of which approximately 75% correspond to triacylglycerols, 18.5% to esters of diterpens and free diterpens and the remaining amount to esters of sterols, free sterols, sterylglucosides, waxes, tocopherols and phosphatides); 25% melanoidins, 3.7–5% minerals and ~6% organic and inorganic acids, and esters (1–4% chlorogenic acids and other phenolic compounds, 1.4–2.5% aliphatic acids and quinic acid and <0.3% inorganic acids), in addition to other minor compounds that may be exclusive to a particular species

(Farah, 2012; Speer and Kölling-Speer, 2017). In this chapter, we will focus on the most common species found around the world, *Coffea arabica* and *Coffea canephora*, which are also the most consumed and the only species studied with respect to human health.

The coffee beverage or brew is an aqueous extract derived from the infusion or percolation of roasted and ground coffee beans, using hot or cold water. The reported amounts of nutrients, main bioactive compounds and other non-nutrients in coffee brews are presented in Table 1. It is worth noting that a coffee brew exhibits very high variability in terms of chemical composition due to many possible variations in raw material production, processing and brewing, which lead to the final product (the brew). The first variable is the blend, which can contain different percentages of coffee beans, each with distinct chemical compositions derived from genetic aspects, origin and degree of maturation (the latter being considered only in the case of a lower quality blend), grown under different conditions and processed via varying postharvest methods. There are also many existing types of roasting profiles and roasting degrees. The roasted beans can then be ground to different sizes and the proportion of powder to water classically used can change dramatically between countries and cultures. For example, whilst in most European countries, the use of 6 g per 100 mL is common for filtered coffees, in Brazil, 10 g or more is used. In Italy, 20 g of ground roasted coffee is also not uncommon for 100 mL. In espresso coffee, although traditionally 6–8 g is used per 25 mL water, an extreme proportion of 10 g per 25 mL water is nowadays often used by third-wave baristas. In addition, there are a variety of brewing methods where pressure, temperature and contact time between the water and the ground coffee vary, and therefore require different amounts of powder. Some methods use filters made up of different materials, which may also influence the composition of the final brew. On top of all this, the size of a cup can vary from about 25 mL for an Italian espresso to 600 mL (20 oz) in the United States. The standard American cup, however, is often reported as being equivalent to approximately 250 mL (8 fluid oz). The traditional European filter cup has been defined in different studies, including that of Floegel et al. (2012), as containing 150 mL. Finally, analytical methods may cause differences in the reported compositional results, especially for the least sensitive and specific methods. Thus, it is understandable that there are no rigorous standard values that represent the chemical composition of a cup of coffee, but rather a range of values. Table 1, therefore, reports the ranges of values found in the literature, but higher or lower values can be found, depending on the compound.

It is worth noting that the yield of a brew (i.e. the amount of solids extracted from the ground coffee found in a cup) may vary from as low as 14% to as much as 60% during soluble coffee production, thus affecting the composition of the brew. Further to this, the extractability of a compound by water will also depend on the amount of soluble solids in the water. Water, containing a high amount of minerals like calcium, magnesium and chloride may extract less solids from the ground coffee, and may also influence the flow time in an espresso machine (Folmer, pers. comm.). Considering that up to 40 g of coffee could be used to prepare 100 mL of a coffee beverage, the amount of soluble solids in coffee brews has been reported to vary from 2 to 6 g per 100 mL cup (Pettraco, 2005) (also referred to as a TC ranging from 2 to 6%). In the traditionally weaker American coffees, measurements of the amount of solids in a few cups yielded 1.2–1.5 g per 100 g. It is worth highlighting that the amount of solids depends upon the degree of roasting.

The main components of the brew are now briefly described.

Table 1 Content range of nutrients, bioactive compounds and other non-nutrients in coffee brews obtained from ground and roasted blends of *C. arabica* species or blends of *C. arabica and C. canephora* species (Trugo, 1985; Macrae, 1985; Clifford, 1985; Urgent et al., 1995; Nunes et al., 1997; Balzer, 2001; Alcázar et al., 2003; Petracco, 2005; Boekschoten et al., 2006; Lang et al., 2010,2011; Farah, 2012; Rubach et al., 2014; Lachenmeier, 2015; USDA, 2017; Glória and Engeseth, 2017)

Nutrients and non-nutrients	Content range[a] (from blends of *C. arabica* or *C. arabica* and *C. canephora* sp.)
Macronutrients	*mg per 100 mL*
Water	94 000–98 500 (TC 1.5–6%)
Simple sugars[b]	0–100 (one report up to 200 mg)
Proteins	120–400
Lipids	180–400
Soluble fibres[c]	200–700 (more commonly between 400 and 500)
Aliphatic acids and quinic acid[f]	692–2140
Vitamins:	
Thiamin (B1)	0.001
Riboflavin (B2)	0.177
Niacin (B3, nicotinic acid)[d]	0.8–10 (more commonly up to 5)
Pyridoxine (B6)	0.002
Folate (B9, DFE)[e]	1
Vitamin C, total ascorbic acid	0.2
Vitamin E (alpha-tocopherol)	0.01
Vitamin K (phylloquinone)	0.1
Tocopherols (α, β, γ)	Traces, only in unfiltered coffees
Minerals: Total ashes	150–500
Potassium, K	115–320
Calcium, Ca	2–4
Sodium, Na	1–14
Phosphorus, P	3–7
Iron, Fe	0.02–0.13
Zinc, Zn	0.01–0.05
Manganese, Mn	0.02–0.05
Bioactive compounds	*mg per 100 mL*
Caffeine	50–380 (commonly between 50 and 150)
Trigonelline	12–50
N-methylpyridinium	2.9–8.7
Diterpenes (cafestol and kahweol)	0.2–1.5 (paper filtered); 2.6–10 (boiled)
Chlorogenic acids	32–500 (commonly 50–150)

Nutrients and non-nutrients	Content range[a] (from blends of *C. arabica* or *C. arabica* and *C. canephora* sp.)
Sum of other phenolic compounds	0.1–0.2
Melanoidins	500–1500
β-carbolins (norharman and harman)	0.004–0.08
Serotonin	0–1.4
Melatonin	0.006–0.008
Polyamines (spermine and spermidine)	0.4
Some undesirable compounds[g]	*μg per 100 mL*
Acrylamide	3.9–7.7
5-hydroxytryptamides[h]	1.2–34.3 (filtered), 350–840 (espresso and French press)
Furan[i]	3.8–262

[a] Content varies with blend, origin, agricultural practices, roasting method and degree, grid, brewing method, amount of coffee powder to water and analytical method.
[b] Arabinose, mannose, galactose, sucrose and minor monosaccharides.
[c] Polysaccharides, mainly galactomannans and type II arabinogalactans.
[d] Daily recommendations for adults:16 mg for men and 14 mg for women (WHO/FAO, 2002).
[e] Dietary Folate Equivalent.
[f] pH 4.3 (acidic coffee, light roast), to 5.8. Common values around 5.0.
[g] These undesirable compounds do not include incidental contaminants.
[h] N-alkanoyl-5-hydroxytryptamides (C5HTs).
[i] Content decreases rapidly after brew preparation.

2.1 Macronutrients

As with other types of plant beverages, coffee brews do not contain excessive amounts of macronutrients (absorbable carbohydrates, proteins and lipids) and calories, unless they are consumed with sugar, milk or cream. According to the USDA National Nutrient Database (2017), 100 mL of filtered coffee (breakfast blend) contains approximately 2 kcal. Exceptions include boiled coffees and similar types of brews, which contain a reasonable amount of lipids as both soluble and insoluble materials are consumed. The nutritional quality of coffee proteins is limited and approximately 50% of this fraction is insoluble and lacks the essential amino acid, tryptophan (Macrae, 1985). In addition, during roasting, most simple sugars and proteins are degraded or changed via the Maillard and pyrolysis reactions and so the amount of protein in the cup is low (120–400 mg per 100 mL, USDA 2017). The detection of up to 200 mg per 100 mL of simple sugars, mainly arabinose, galactose and mannose, and lower amounts of sucrose, fructose and glucose, has been reported for brewed coffee and dissolved soluble coffee at 2% (coffee/water), with the latter being higher (Macrae, 1985; Trugo, 1985). Galactomannans and type 2 arabinogalactans are considered to be the predominant soluble dietary fibres in a coffee beverage (often between 140 and 650 mg per 100 mL), of which approximately 70% is comprised of galactomannans (Gniechwitz et al., 2007; Farah, 2012); however, care is needed to distinguish polysaccharides from total carbohydrates in order to avoid overestimation.

2.2 Micronutrients

With regards to micronutrients, the brew may contain a reasonable amount of vitamins and minerals. Niacin, in the form of nicotinic acid, is the main vitamin in a coffee brew and is also known as vitamin B3 or PP (pellagra preventing), and participates as a coenzyme in various metabolic processes involved in energy metabolism and tissue health. Additionally, it contributes to the normal functioning of the nervous system, among many other functions. Deficiency of this vitamin causes pellagra, a disease characterized by skin lesions and diarrhea, among other symptoms (Arauz et al., 2015). Regular coffee consumption can supply an essential part of the daily recommendation for adults, which is 16 mg for men and 14 mg for women according to FAO standards (WHO/FAO, 2002). Niacin can also be produced in the liver from a reasonable amount of tryptophan (60mg thyptophan makes 1mg niacin), which is present in animal proteins, nuts, and some other seeds. The intake of niacin from coffee is particularly important in places where tryptophan consumption is low, such as in the rural and less developed areas of Central America and Central Africa where corn (which is poor in niacin and tryptophan) is the staple food (Carpenter, 1983; Macrae, 1985). However, it is worth noting that people who consume corn as tortillas or similar foods made with corn flour pre-treated with alkaline water are not at risk of niacin deficiency as those who consume untreated corn and corn flour (Carpenter, 1983). Most recently, supplementation of nicotinic acid has been used to decrease low and very low density lipoproteins (LDL and VLDL) levels (Le Bloch et al., 2010) and to contribute to the hepatoprotective effect of coffee (Arauz et al., 2015) and therefore, an additional food source of niacin, like coffee is welcome considering that excess amounts of the vitamin are excreted in urine (Wang et al, 2001).

Coffee brews may also contain very small amounts of nicotinamide, another form of niacin, and other B vitamins (thiamin, riboflavin, pyridoxine, folic acid), ascorbic acid (vitamin C), and phylloquinone (vitamin K) (Macrae, 1985; USDA 2017). Unfiltered coffees can contain small residual amounts of tocophenols (α, β and γ – the latter two predominate), although during and after roasting almost all of the original amount in green coffee (about 60 mg/100g) is degraded or oxidized (Macrae, 1985).

Considering the different factors that create a large variability in the composition of roasted coffee, in addition to differing brewing methods and the extractability of different minerals, the values of total ash ranging between 150 and 500 mg per 100 mL have been reported, including data from dissolved soluble coffee (containing 7–10% ash), prepared at 2% (ground coffee to water). For infusions prepared from ground coffee in Poland (Grembecka et al., 2007), the consumption of 300 mL (2 cups) was estimated to supply, on average, 4.5% of the recommended dietary allowance (RDA) for Mg, 3.5% for K, 2.8% for Mn, 2.4% for Cr, 1.9% for P, 0.32–0.43% for Ca and Na, 0.26–0.33% for Cu and Fe, 0.13% for Zn and 2.6–15.6% for Ni. Higher average percentages were estimated from the ingestion of a 300-mL beverage prepared from instant coffee: 12.3% for Mg, 8.9% for K, 8.6% for Mn, 4.9% for Cr, 7.4% for P, 1.6% for Ca, 2.5% for Na, 0.32% for Cu, 2.9% for Fe, 0.35% for Zn and 4.9–29.7% for Ni. Estimates indicate that the consumption of coffee in typical amounts does not exceed the tolerance limits for the ingestion of toxic metals, such as Pb and Cd.

In a study performed in Portugal (Oliveira et al., 2012), two cups of instant coffee (total of 4 g) were estimated to supply 9.5% of the RDA for K, 5.2% for Mg, 4.4% for Mn, 3.5% for Ni, 2.2% for P, 1.5% for Fe, 0.5% for Cr, 0.4% for Ca and 0.2% for Na. In a more recent study, also performed in Portugal (Oliveira et al., 2015), one cup of an espresso coffee beverage (prepared from 5 to 6 g of ground coffee) was estimated to provide 5.2–7.0% of the RDA for

K, 2.8–7.2% for P, 1.4–2.2% for Mg, 1.4–1.9% for Mn, 0.14–0.28% for Ca, 0.07–0.15% for Fe and <0.02% for Na.

Other similar studies indicate that regular coffee intake (300 mL) does not contribute, in most cases, to amounts higher than 10% of the daily recommendations in different countries.

2.3 Bioactive compounds

It is well known that nutrients are required to maintain normal body functions and are therefore bioactive. However, the term 'bioactive compounds' commonly refers to minor food constituents that exert biological functions other than nutritional functions. These compounds are commonly found in plants, and in a few animals that feed on them, and their chemical structures and biological functions vary widely.

Although most health-related aspects of coffee have been attributed to the beverage and not to individual compounds, mechanistic studies have suggested that some specific bioactive compounds play key roles as co-adjuvant agents in disease prevention. In addition to caffeine, the most studied bioactive compounds of coffee are chlorogenic acids and their lactones, trigonelline and their derivatives, the diterpenes, cafestol and kahweol and melanoidins. The polysaccharides, galactomannans and type II arabinogalactans, and β-carbolines are amongst the emerging bioactive compounds for which there is still insufficient information to substantiate any health effects. Also, relatively recently, it has been suggested that some coffee amines exert positive effects on health when consumed in low quantities, which are referred to as bioactive amines. Each of these compounds or group of compounds will be introduced hereafter and their effect on health will be briefly discussed. The main undesirable compounds in coffee are also introduced.

2.3.1 Caffeine

Caffeine is the most well-studied compound present in coffee, and its mechanisms of function are generally well documented. First, there are psychostimulating effects which include an acute impact on mental performance as well as long-term influence on the risk of developing neurodegenerative diseases such as Parkinson's and Alzheimer's. Caffeine has also been found to improve physical performance, which will be discussed in the next section on the health effects of coffee. Most recently, a number of studies have reported new bioactive effects for caffeine, and one of the emerging effects is the enhancement of the antioxidant effect of coffee. Caffeine metabolites, especially 1-methylxantine and 1-methylurate, have exhibited an antioxidant activity *in vitro* (Moura-Nunes et al., 2009). Corroborating these results, the average plasma iron-reducing capacity of human subjects after regular coffee consumption was higher than that recorded after the consumption of decaffeinated coffee, suggesting that whole coffee is more efficient than decaffeinated coffee with respect to its antioxidant capacity (Moura-Nunes et al., 2009).

There are a few *in vitro* studies showing that caffeine contributes to the antibacterial effect of coffee against *Streptococcus mutans*, a significant contributor to cariogenic bacteria, as well as intestinal pathogenic bacteria (Antonio et al., 2010). Over the last few years, it has also been suggested that caffeine exerts an antihyperlipidemic effect (decreased storage of triglycerides and cholesterol) by inhibiting lipogenesis and stimulating lipolysis through the regulation of the gene expression responsible for lipid metabolism in liver cells (Quan, 2013). These are just a few of the various emerging effects of caffeine.

2.3.2 Chlorogenic acids and their derivatives

Chlorogenic acids and their derivatives – Chlorogenic acids are the main phenolic compounds in coffee and this group of compounds includes approximately ten major esters and four major lactones (produced during roasting), in addition to dozens of trace compounds. The total amount of chlorogenic acids in *C. canephora* beans is almost double than that found in *C. arabica* beans, and because chlorogenic acids are partly degraded or transformed during roasting, dark roasted coffees contain lower amounts of these compounds. In roasted products, the difference between species is significantly reduced.

These compounds are frequently referred to as powerful antioxidants and anti-inflammatory compounds due to the results of *in vitro* and animal studies, as well as a few human studies (Santos et al., 2006; Torres and Farah; 2016, Folmer et al., 2017), but the mechanism of action of the different compounds and how they are related to the prevention of the disease remains, to a large extent, unknown.

Owing to the high concentration of chlorogenic acids in coffee brews (Table 1), compared with chlorogenic acids and other phenolic compounds in foods, in general, they may play a major role in the diet of consumers as a source of antioxidative compounds. The significant contribution of chlorogenic acids to the dietary intake of antioxidative compounds is exemplified in a number of reports from different countries in which, based on their official food consumption database or other types of surveys, coffee was the main contributor to total dietary antioxidant capacity, that is, Brazil (66%), Norway (64%), Italy (38% for women and 27% for men), Spain (45%), Japan (56%) and the Czech Republic (54.6% for women and 43.1% for men) (Torres and Farah, 2016). However, it should be kept in mind that the intake of chlorogenic acids from coffee does not replace the intake of antioxidants from other foods, as each has its own specific bioactivity.

Together with other polyphenols, carotenoids and additional classes of antioxidative compounds, chlorogenic acids and their lactones have been associated with a decrease in the risk of Alzheimer's and type 2 diabetes, amongst various degenerative diseases (Kasai et al., 2000; Obuleso et al., 2011; Farah, 2012).

Long before epidemiological studies investigated the association between coffee consumption and health effects, the antimutagenic property of chlorogenic acids and their metabolites was discovered. Recent studies have confirmed these findings and elucidated several mechanisms of chemopreventive action, which include modulating the expression of the enzymes that are involved in endogenous antioxidant defences, DNA replication, cell differentiation and ageing (Feng et al., 2005; Ramos, 2008;Jurkowska, 2011), metal chelation, inactivation of reactive compounds and metabolic pathway changes (Kasai et al., 2000; Farah, 2012). In the colon, for example, chlorogenic acids may inactivate free reactive radicals and as a result help prevent colon cancer (Ludwig et al., 2014).

Additional health effects observed *in vitro* and in animal studies include hepatoprotective (including cirrhosis, liver cancer and other liver diseases), immune-stimulatory and antibacterial and antiviral activities. Synthetic derivatives of these compounds have also inhibited HIV-1 replication in cells, which could play a role in research towards drugs that inhibit HIV (Farah, 2012). Additionally, recent *in vitro* studies have suggested that after coffee consumption, the unabsorbed portion of chlorogenic acids may serve as a substrate to stimulate the growth of beneficial intestinal bacteria; however, this effect requires further investigation due to conflicting data in different studies (Sales et al., 2017).

Trace amounts of other phenolic compounds, that is, isoflavones, proantocyanidins and lignans, have been identified in coffee (Farah and Donangelo, 2006), which possibly enhance the beneficial effects of chlorogenic acids.

2.3.3 Melanoidins and polysaccharides

Melanoidins and polysaccharides – Coffee melanoidins have gained importance over the years because of their contribution to, amongst others, the antioxidant and antimicrobial effects of coffee (Rufián-Henares and Morales, 2007). This is at least, in part, due to the incorporation of chlorogenic acids and other bioactive compounds into their structure during roasting (Farah, 2012).

As melanoidins are not digested, they may act, in combination with coffee polysaccharides (mainly galactomannans and type II arabinogalactans), as soluble dietary fibres. They are largely indigestible and thus fermented in the gut (Borrelli et al., 2004; Gniechwitz, 2008). A recent study concluded that the consumption of 0.5–2 g melanoidins per day (present in 2–5 cups) contributes up to 20% of the recommended 10 g of daily soluble dietary fibre intake. It has also been hypothesized that these substances may stimulate the growth of beneficial bacteria in the lower digestive tract (Fogliano and Morales, 2011), in the same way as chlorogenic acids; however, the data remain controversial.

As with chlorogenic acids (Passos et al., 2014), it has been hypothesized that melanoidins can enhance immune-stimulating properties and contribute significantly to reducing the risk of colon cancer (Vitaglione et al., 2012; Moreira et al., 2015; Fogliano and Morales, 2011), which might occur in different ways: i) by increasing the elimination rate of carcinogens through higher colon motility and faecal output, ii) by decreasing colon inflammation through improved microbiota balance (prebiotic effect) and iii) by serving as a 'sponge' for free radicals in the gut (Garsetti et al., 2000; Folmer et al., 2017).

2.3.4 Trigonelline and derivatives

Trigonelline and derivatives – Trigonelline is another compound that has gained importance in recent years due to its potential contribution to the protective effect of coffee against diseases. *In vitro* and animal studies have reported different involvements of trigonelline against type 2 diabetes (Yoshinari and Igarashi, 2010), as well as neuroprotective (Hong et al., 2008; Tohda et al., 2005), antitumour (Hirakawa et al., 2005) and phytoestrogenic effects (Farah, 2012). Beans from *C. arabica* species contain higher amounts of this compound, compared with *C. canephora*, and as with chlorogenic acids, trigonelline undergoes changes and degradation during roasting; hence, dark roasted coffees contain low amounts. However, 10–20% of the original amount of trigonelline is converted into nicotinic acid (niacin) (Farah, 2012). In addition to its vitamin function, niacin is also involved in other bioactive functions, presenting antidiabetic (Yoshinari and Igarashi, 2010), antioxidant and hepatoprotective (Arauz et al., 2015) effects *in vitro* and in animal studies. The compound *n*-methylpyridinium and other pyridinium derivatives are additional thermal degradation products generated by the decarboxylation of trigonelline. It has been reported that like trigonelline, *n*-methylpyridinium promoted higher glucose utilization in liver cells, stimulating cellular energy metabolism and contributing to the protective effect against type 2 diabetes (Riedel et al., 2014). Pyridinium derivatives have also been reported to present antioxidant/chemopreventive (Somoza et al., 2003), hepatoprotective (Gebicki et al., 2008), vasoprotective (Lang et al., 2011) and antithrombotic effects (Kalaska et al., 2014).

2.3.5 Diterpenes

Diterpenes – Cafestol and kahweol are diterpenes present in coffee mainly in the form of salts or esters of (predominantly) saturated and unsaturated fatty acids. They represent approximately 20% of the lipid fraction of coffee, with cafestol being more abundant. Higher levels of diterpenes are found in *C. arabica* than in *C. canephora* species. Coffee diterpenes exhibit strong anticarcinogenic and hepatoprotective properties *in vitro* (Farah, 2012).

Diterpene levels in the coffee cup vary significantly based on the natural variations in green coffee beans, roasting conditions and preparation methods (Urgert et al., 1997; Gross, 1997;Urgert and Katan, 1995). Whilst filtered and soluble coffees are practically diterpene-free (due to their poor solubility in water, they are trapped by paper filters), espresso-based methods contain higher levels of diterpenes, which are, on the other hand, significantly lower than those found in French press or Turkish coffee (2–10 mg per cup) (Farah, 2012).

2.3.6 β-carbolines

β-carbolines – These are alkaloids formed in coffee mainly during roasting and the two identified β-carbolines in coffee are norharman and harman. Despite some past controversies regarding the neurological and toxicological effects of these compounds in studies using high doses in animals, β-carbolines have been recently associated with potentially positive effects, including neurological ones, with antidepressive and neuroprotective properties. It has also been suggested that they may reduce the risk of diabetes. The total concentration of these compounds in the brew is highly variable in the literature, from 4 to 80 μg per 100 mL, but typical concentrations are reported to be in the range of 4–20 μg per 100 mL, being primarily dependent on the coffee species. Roasted *C. canephora* beans have consistently higher amounts of β-carbolines than *C. arabica* beans (Farah, 2012;Rodrigues and Casal, 2017;Casal, 2017).

2.3.7 Bioactive amines

Bioactive amines – These compounds are organic bases with psychoactive, neuroactive or vasoactive activity and participate in a number of processes in the human body. Coffee amines that present positive health effects in *in vitro* and in animal studies, other than their known physiological effects in the body, are called bioactive amines. However, no direct association between their presence in coffee and benefits to human health has been found. The main bioactive amines are the indolamines, serotonine and melatonin, and the polyamines, spermine and spermidine. Mean reported concentrations of serotonin in coffee beverages vary from non-detected to 90 μg per 100 mL in most coffee beverages, but from 372 to 1354 μg per 100 mL in Turkish coffee. Information regarding melatonin is rare; however, levels ranging from 6 to about 8 μg per 100 mL have been found. Although serotonin is a neuroactive substance with various positive effects on well-being, serotonin from the diet cannot cross the blood–brain barrier and can only be produced in the brain; however, serotonin from the diet can have other potentially relevant roles including broncho- and vasoconstrictor, antihypertonic, antioxidant and antiallergic and antidiuretic effects. It can also help to modulate the volume and acidity of gastric juice. Spermine and spermidine (reported amounts for each vary from non-identified to more than 150 μg per 100 mL brew) are efficient free radical scavengers in several chemical and enzymatic systems. Other amines have also shown positive effects on health when in low quantities (Gloria and Engeseth, 2017).

2.4 Undesirable compounds in coffee

A few compounds derived from microbial contamination (ocratoxin A, biogenic amines), pesticides or chemical reactions that occur during the roasting process (mainly acrylamide, furan and polycyclic aromatic hydrocarbons – PAH) have been of concern to health authorities such as the Food and Drug Administration (FDA), Food and Agriculture Organization of the United Nations (FAO) and the European Food Safety Authority (EFSA).

2.4.1 Ocratoxin A

Ocratoxin A and similar toxins are derived from green bean contamination with mould and can be avoided by carefully harvesting, processing and storing coffee, which is reflected in good quality. Ocratoxin A is gradually degraded by the high temperatures of roasting and its residual levels in roasted coffee are regulated in many countries. In the European Union, regulation 1881/2006 states that for roasted coffees, the maximum limit is 5 μg per kg.

2.4.2 Pesticides

Pesticides comprise a large number of substances belonging to different chemical groups, which are used to control plant diseases, pests or weeds. They can be neurotoxic or inhibit vital metabolic reactions in living beings, targeting different mechanisms, and individual pesticides present different levels of toxicity. In order to protect human safety and health resulting from pesticide application during coffee production, many countries have put these chemicals under strict legislation and surveillance. For example, in the United States, the FDA establishes the maximum amount of a pesticide allowed to remain in food, as part of the process of regulating pesticides. Presently, tolerance limits for about 43 pesticides in coffee are listed by the FDA. In Japan, the maximum tolerated residual limits are amongst the lowest (often 0.01 ppm). Despite all the regulations, pesticide residues have been found via analyses of green coffee performed prior to importing at differing occasions and in different countries. Due to restrictions and monitoring, the residual levels of pesticides in commercial ground and roasted coffees are usually very low and within the established limits. The solubility of residual pesticides after roasting is often low and therefore low amounts are found in the brew, especially in filtered coffee. However, the toxicity of their metabolites in seeds or degradation products during roasting has not been well studied and could be higher than that of the pesticides themselves (Farah, 2012; Cunha and Fernandes, 2017).

2.4.3 Acrylamide, furan and PAH

Acrylamide, *furan and PAH* are derived from reactions that occur during roasting, more specifically Maillard and pyrolysis (Farah, 2012), which can occur in many other heated foods such as French fries or bread. Studies assessing the risk of their concentrations in brews have not found considerable amounts which could cause harm to human health, as epidemiological studies have failed to find a link between these compounds in coffee and an elevated risk of cancer or other diseases (Lipworth et al., 2012; Nkondjock, 2012).Amongst all these compounds, acrylamide is the most abundant and the one which many food and health authorities have been most concerned about. It is formed at the beginning of the roasting process and its levels decrease thereafter to a certain degree (Farah, 2012). Acrylamide has been associated with cancer in one study using laboratory rodents which were exposed to extremely high concentrations (1000–10 000 times physiological ranges) (Mucci and Adami, 2009). Whilst the US FDA (FDA, 2016) reported that coffee is a significant source of acrylamide exposure for adults, the EFSA's recent opinion on acrylamide stated that health authorities do not see any direct evidence of cancer risk (EFSA 4104, 2015b; Folmer et al., 2017). The

European Commission's recommended indicative value is 450 μg acrylamide per kg of roast and ground coffee, a level which is generally achievable for commercial products.

Furan, hydroxymethylfurfural (HMF) and furfural are heterocyclic, low molecular weight molecules with a furan ring and potential carcinogenicity in common. Furan has been classified as a possible carcinogen (Group 2B) by the International Agency for Research on Cancer (IARC). Group 2B status is assigned to compounds and exposure conditions for which there is limited evidence of carcinogenicity in humans and insufficient evidence of carcinogenicity in experimental animals, and therefore requires further investigation. No human studies are available regarding the effects of furan and there is a significant uncertainty in the extrapolation of risk from animal assays performed in the laboratory to the equivalent risk for humans. When dealing with furfuryl and furan derivatives, the EFSA report (EFSA, 2011) concluded that notwithstanding some indications of *in vitro* genotoxicity, based on available data of exposure and on *in vivo* genotoxicity studies, which gave negative results for the carcinogenicity evaluation in rats and mice, under normal conditions, these compounds are of no concern to human health (Folmer et al., 2017). Regulatory bodies, including the US FDA and the EFSA, have made no recommendations regarding the maximum levels of furan in the dietary intake.

In coffee, these compounds are formed during the roasting stage mainly via thermal degradation/Maillard reaction of reducing sugars, alone or in combination with amino acids or via the thermal degradation of amino acids. The consumed levels of furan in coffee are highly variable and reflect not only the preparation methods but also the roasting conditions; however, these compounds are not exclusive to coffee. The major contributors to furan exposure in adults and teenagers were estimated to be fruit juice, and milk-based and cereal-based products. Additionally, jarred baby foods were also major contributors in toddlers (Ferreira et al., 2017).

Examples of a range of furan concentrations, obtained using coffees from the Spanish market prepared by different methods, are 12–146 μg per litre with the lowest values found in instant and filtered coffee and the highest values found in espresso. Boiled coffee was not evaluated. The concentration for commercially packed coffee capsules was approximately 240 μg per litre. The furan content of coffee brews from automatic coffee vending machines ranged from 11 to 262 μg per litre; however, this is not representative of what people would consume. Due to its high volatility, after coffee preparation, losses occur rapidly upon mixing and waiting for the brew to cool down (Ferreira et al., 2017).

The most important contributors of HMF in the diet are dried fruits, caramel, vinegar, bread and coffee. The content of HMF in coffee samples from coffee vending machines (8 g of ground coffee to 100 mL water) ranged between 4 and 60 mg per litre with a mean content of 28.8 mg per litre. The concentrations of furfural in coffee brews from European vending machines ranged between 0.30 and 1.30 mg per litre (Ferreira et al., 2017). To date, no measures have been identified to mitigate furan without impacting the typical coffee aroma.

PAH, from which benzo[a]pyrene is the most relevant from a toxicity point of view, can be formed in coffee and other foods that are severely roasted or exposed to very high temperatures. This compound is classified by the IARC as probably carcinogenic to humans. However, the level of exposure to PAH from coffee is low and within the safe limits set by International Agencies (IARC, 2010; EFSA, 2008).

2.4.4 Biogenic amines

Biogenic amines are organic bases of low molecular weight that participate in the regular metabolic processes of plants, microorganisms and animals. They are produced in the body and can also be provided by the diet and from the microbial flora of the intestine; at high concentrations, however, they can pose a toxicological risk.

Examples include histidine, tyramine, tryptamine, cadaverine and putrescine. Histidine is the most toxic and is associated with a hypotensive effect and headaches. Putrescine, cadaverine and tyramine seem to be toxic in higher doses in animals, but the individual sensitivity to these compounds in humans varies considerably and causes different responses. In coffee, biogenic amines originate from the action of microbial enzymes on amino acids during fermentative processes, suggesting inappropriate storage or low-quality defective fermented seeds. Roasting may deconjugate and increase the amount of some free biogenic amines, whilst most other amines are degraded (Gloria and Engeseth, 2017b; Farah, 2012).

3 Main beneficial health effects of coffee

The role of coffee drinking for the purpose of socialization and relaxation is indirectly important for health, as stress plays a major role in the development of several diseases that may lead to serious complications and death. Additionally, coffee has been shown to help prevent degenerative disorders, many of which are related to neurostimulating, antioxidant and anti-inflammatory effects. A prospective US cohort study (Freedman et al., 2012) examined the association of coffee drinking with subsequent cause-specific and total mortality in the National Institutes of Health– AARP Diet and Health Study. This study involved more than 400 000 people and is, so far, the largest human study investigating coffee and health. A significant inverse association between coffee and specific deaths due to heart disease, respiratory disease, stroke, injuries and accidents, diabetes and infections was found (all of which are amongst the 10 leading causes of death, WHO, 2017–Fig.1). Total mortality was reduced considerably by up to 16% for both men and women who drank 4–5 cups of coffee a day. Similar associations were observed whether participants drank predominantly caffeinated or decaffeinated coffee (Folmer et al., 2017). Although scientific studies can link certain compounds to specific mechanisms, it is likely that most contributions to decreasing the risk of certain diseases are caused by synergistic or additive effects with various compounds present in coffee. The next section presents summaries of research results on the effect of coffee and health, exploring the most studied effects individually.

3.1 Coffee consumption for mental and physical performance and well-being

The stimulating effect of coffee is well known and is due to caffeine's ability to enhance mental performance, which includes enhancing alertness and perception (Einother and Giesbrecht, 2012). According to the EFSA, who reviewed existing evidence of caffeine on mental performance (EFSA 2045, 2011c), generally, a dose of 75 mg is required to obtain these effects, although very large differences in individual responses to caffeine are observed. Caffeine consumption can also improve other functions such as memory (Nehlig, 2010; Borota et al., 2014) and mood (Smith, 2005; Olson et al., 2010). Coffee components other than caffeine have also been shown to influence cognitive performance in an elderly population, though to a smaller extent than caffeine. Decaffeinated coffee enriched in chlorogenic acids can improve alertness and reduce headaches and mental

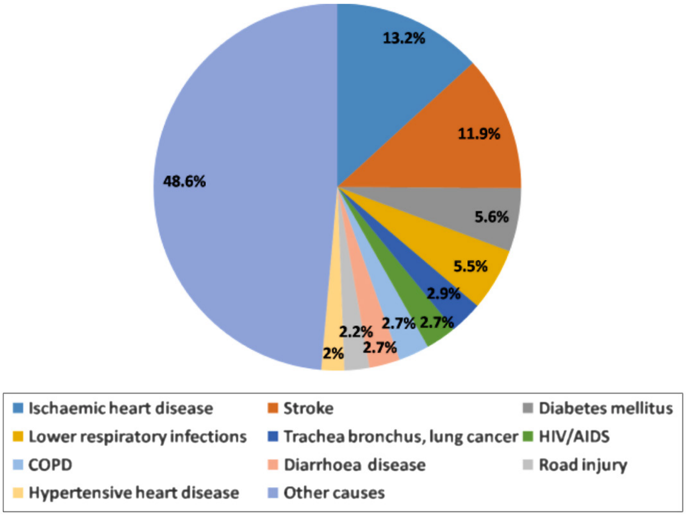

Figure 1 The ten leading causes of death in the world by percentage (data from WHO, 2017, updated in 2014).

fatigue in comparison to non-enriched decaffeinated coffee. The effects may be partly attributed to chlorogenic acids, but other compounds naturally present in coffee are also suggested to play a role (Camfield et al., 2013; Cropley et al., 2012; Folmer et al., 2017).

The large inter-individual variability of the stimulating effects of caffeine is due to the difference in the ability to metabolize and eliminate it from the body. Whilst for most people, it takes about three to six hours to eliminate 50–75% of the caffeine and its metabolites (Goldstein, 2010), for some people, it can take much longer. The effects of several cups of coffee on these individuals, usually called 'slow metabolizers', may therefore be accumulative for a while. The variability in the enzymatic breakdown of caffeine may account for its variable effect on sleep induction and arousal (Youngberg et al., 2011). The stimulating effects of caffeine tend to be stronger when the individual is in a state of fatigue or in elderly people (van Boxtel and Schmitt, 2004). However, habitual caffeine consumers may suffer less from these issues as they develop a tolerance. In this case, caffeine will, for example, still disrupt their sleep, but to a lesser extent than for people who are not habitual consumers (Childs and de Wit, 2012; Drapeau, 2006). It is the responsibility of each person to pay attention to his or her response to caffeine intake at different times of the day, and adapt intake patterns accordingly.

Coffee and other caffeine vehicles have been used by athletes for a long time, and the initial papers discussing the mechanisms involved date back to 1978 (Costill, 1978). More specifically, caffeine exerts a positive effect on the endurance and exercise capacity, due to the effect on neural mechanisms (Spriet and Gibala, 2004;Folmer et al., 2017). Caffeine also seems to reduce the pain perception due to an increase in the secretion of β-endorphins which exhibit analgesic properties (O'Connor et al., 2004). It is well documented that caffeine can enhance endurance and coordination, stop–go events (e.g. team and racket sports) and sports involving sustained high-intensity activity lasting from

1 min up to an hour (e.g. swimming, rowing and running races) (Jenkins, 2008; Hogervost et al., 2008; Folmer et al., 2017).

Based on the scientific studies, the active dose of caffeine was found to be 3 mg per kg of bodyweight, to be taken 1 h before exercise (EFSA, 2054, 2011c; Goldstein et al., 2010). For a person weighing about 70 kg, this amount would thus be equivalent to 210 mg. According to the EFSA caffeine safety report (EFSA 4102, 2015a), it is safe to consume single doses of 200 mg of caffeine less than 2 h prior to intense exercise. However, the amount and time prior to exercise for an optimal effect will vary for different individuals due to differences in metabolic rates (Folmer et al., 2017). In 1994, caffeine was removed from the list of banned substances.

The impact of caffeine on the mental and physical health of women and children is a more recent area of interest, even if the initial papers appeared in the early 1990s. Evidence suggests that the normal hormonal changes during pregnancy slow the body's ability to metabolize caffeine. Therefore, a given dose of caffeine can have longer-lasting effects (as long as 15 h in the third trimester) (Kuczkowski, 2009). Even though the EFSA report on caffeine safety (EFSA 4102, 2015a) concludes that its consumption is safe for pregnant and lactating women, it recommends an intake reduction to a maximum of 200 mg throughout the day. Based on scientific findings, there is no risk of adverse birth weights for caffeine consumption below these values. Nevertheless, the risks of very high intake (more than 600 mg of caffeine per day) include foetal growth retardation and low weight for gestational age (Sengpiel et al., 2013). Although there is no consensus in studies suggesting that caffeine could delay the time of conception, it may be prudent for women who have difficulty in conceiving to limit the caffeine intake to less than 300 mg per day (Higdon and Frei, 2006; Folmer et al., 2017).

It is known that caffeine is present in the milk of lactating coffee drinkers with a peak appearing about 1 h after consuming a caffeinated beverage (Stavchansky et al., 1988; Nehlig and Debry, 1994). For this reason, doctors recommend that breastfeeding women keep caffeine intake below 200 mg per day (EFSA 4102, 2015a). At these levels, studies show that the sleep time of nursing infants is similar to controls (Santos et al., 2012; Clarc and Landholt, 2016; Folmer et al., 2017).

When it comes to children and coffee consumption, there are major cultural differences in both overall coffee consumption and consumption guidelines. For example, in most European countries, habitual coffee consumption starts when children become adults, and until the age of 10, chocolate and tea are the main sources of caffeine (EFSA 4102, 2015a). Brazil has implemented an active coffee school programme based on the findings that 20% coffee added to a glass of whole milk helps children perform better in school (ABIC, 2016). Additionally, there are studies that show that caffeine may attenuate the symptoms of attention-deficit syndrome (Garfinkel et al., 1981). European adolescents consume less coffee and their source of caffeine intake is widely distributed amongst different types of food and beverages (EFSA 4102, 2015a). In the United States, this section of the population primarily consumes caffeine from soft drinks (USDA, 2015).

As information on the impact of caffeine on the health of children and adolescents is scarce, it is difficult to derive general conclusions on safe intake levels. Caffeine doses of about 1.4 mg per kg bodyweight or more may impact sleep quality in adults, particularly when consumed close to bedtime (EFSA 4102, 2015a). For this reason, and because data on safe habitual caffeine intake for children and adolescents are insufficient, the EFSA suggests a limit of 3 mg of caffeine per kg of bodyweight per day (EFSA 4102, 2015a), which would equal around 90 mg for a 10-year old (Folmer et al., 2017).

Canadian authorities are more conservative and suggest a limit of 2.5 mg per kg of bodyweight per day (Health Canada, 2011). The short-term risk associated with children and caffeine consumption is that caffeine may cause anxiety and nervousness (Nawrot et al., 2003).

3.2 Coffee and cognitive health

The acute effects of caffeine were discussed in the mental performance section. In this section, we will look at the long-term effects of coffee on reducing the risk of cognitive degenerative diseases. Cognitive functions such as verbal ability, inductive reasoning and perceptual speed decrease after 20 years of age. Genetics, life events and lifestyle factors impact the rate and amplitude of this decline (Hedden and Gabrieli, 2004; Folmer et al., 2017). A large number of epidemiological studies relate the regular consumption of coffee to a reduced appearance of cognitive decline in the elderly (Arab et al., 2013; Ritchie, 2007; Corley, 2010). A meta-analysis of these human studies suggests that there is a clear protective effect of caffeine consumption, rather than from coffee itself (Santos et al., 2010; Ryan, 2002).

Alzheimer's disease is the most frequent cause of dementia, leading to a progressive cognitive decline. Whilst there is currently no medication for Alzheimer's disease (Waite, 2015), there are studies that show an inverse association between the coffee consumption and the development of Alzheimer's disease, with a 27% risk reduction (Barranco Quintana et al., 2007; Waite, 2015; Folmer et al., 2017). The mechanism is believed to be related to the anti-inflammatory effect of caffeine on the A1 and A2 receptors, in addition to reducing the deposits of toxic beta-amyloid peptide in the brain, a pathological characteristic in patients with Alzheimer's disease (Rosso, 2008; Arendash and Cao, 2010). In addition to caffeine, the intake of polyphenols also seems to help decrease the risk of Alzheimer's disease. Emerging evidence from animal models also links chlorogenic acids to the prevention of neurodegenerative disease and ageing (Esposito et al., 2002; Ramassamy, 2006). Although the involvement of coffee polyphenols in the human cognitive function has not been well studied, the number of findings on the *in vitro* neuroprotective effects of polyphenols in general is rapidly increasing (Lakey-Beitia, et al., 2015). Initial indications relate the anti-inflammatory effects of polyphenols to the reduced risk of developing Alzheimer's disease. Other proposed mechanisms could be i) inhibition of the enzymes acetylcholinesterase and butyrylcholinesterase in the brain, as this retards acetylcholine and butyrylcholine breakdown and ii) the prevention of oxidative stress–induced neurodegeneration due to its high antioxidative activity (Oboh et al., 2013, Folmer et al., 2017).

Similar to Alzheimer's disease, a large number of epidemiological studies have reported an inverse relationship between the caffeine consumption and the risk of developing Parkinson's disease. The latter is a neuropathological disorder that slows down the motor function, whilst generating resting tremors, muscular rigidity, gait disturbances and impairing postural reflex. It involves the degeneration of neurons in the brainstem (Kuwana et al., 1999). Coffee consumption appears to reduce, or delay, the development of Parkinson's disease. From the meta-analysis of 26 studies, a 25% lower risk of Parkinson's disease was found in coffee drinkers compared with non-coffee drinkers. The mechanism is probably related to the capacity of caffeine to block the A2 adenosine receptors in the brain (Costa et al., 2010). Studies recently outlined a possible additional mechanism. A rodent model showed that trigonelline may exert a neuroprotective effect, inducing a significant reversal of motor dysfunction (Nathan et al., 2014; Folmer et al., 2017).

3.3 Coffee and cardiovascular disease

One of the misinterpretations linking coffee and health stemmed from the belief that the risk of cardiovascular disease, the leading cause of death in the world according to the WHO (13.2% and 2% of deaths due to ischaemic and hypertensive cardiovascular diseases, respectively, WHO, 2017;Fig.1), was increased by drinking coffee. This belief was supported by the fact that caffeine increases blood pressure and acutely reduces insulin sensitivity after coffee consumption. However, it is now known that most acute caffeine effects cease to exist with regular coffee consumption due to adaptation mechanisms and that other coffee components, mainly chlorogenic acids and trigonelline, have compensatory effects on endothelial dysfunction and insulin resistance. Additionally, *in vitro* and animal studies indicate that coffee has high antioxidant and anti-inflammatory potential, improves endothelial dysfunction and reduces insulin resistance, which are key mechanisms for cardiovascular protection (Rebello and Van Dam, 2013).

Corroborating these findings, dozens of studies have shown the inverse association between coffee consumption and cardiovascular diseases. Andersen et al. (2006) studied the relationship of coffee drinking with total mortality and mortality attributed to cardiovascular disease, cancer and other diseases with a major inflammatory component. A total of 41 836 postmenopausal women aged 55–69 years at baseline were followed for 15 years. During this period, there were 4265 deaths. Evaluating the causes of mortality, the authors observed that coffee consumption increasingly reduced the risk of cardiovascular and other inflammatory diseases in postmenopausal women, thereby decreasing mortality from these diseases. This effect was attributed to the ability of coffee to inhibit inflammatory processes via its antioxidative and anti-inflammatory compounds.

A meta-analysis was carried out by Crippa et al. (2014), using 21 prospective studies, with 997 464 participants and 121 915 reported deaths. Results indicated that coffee consumption is inversely associated with all-cause and cardiovascular disease mortality and that the risk was increasingly reduced for those who consumed 3 to 4 cups. Similar results were observed by Malerba et al. (2013).

Another recent study by Ding et al. (2015) examined the causes of death of 19 524 women and 12 432 men from two large cohort studies in the United States, the Harvard Health Professionals Follow-up Study and the Nurses' Health Study (1 and 2). Inverse associations were observed between the consumption of regular and decaffeinated coffee and the deaths due to cardiovascular and neurological diseases. When restricting to those that had never smoked, the all-cause mortality risk was also increasingly reduced as the number of cups increased; however, higher consumption reduced the benefit somewhat.

Stroke is the second leading cause of death in the world as estimated by the WHO (11.9% of the total deaths, WHO, 2017; Fig.1); however, data on the association between coffee consumption and risk of stroke are scarce. A study by Lopez-Garcia (2009) analysed data from a prospective cohort of 83 076 women in the Nurses' Health Study for 24 years. Results evidenced that long-term coffee consumption is not associated with an increased risk of stroke in women. In contrast, data suggested that coffee consumption may modestly reduce this risk. Decaffeinated coffee was associated with a trend towards a lower risk of stroke after adjustment for caffeinated coffee consumption. Using data from 11 prospective studies with 479 689 participants and 10 003 cases of stroke, a meta-analysis performed by Larsson and Orsini (2011) corroborated the results obtained for women, finding an inverse, although modest, association between moderate coffee consumption and risk of stroke.

3.4 Coffee and type2 diabetes

Diabetes mellitus is characterized by a high blood glucose level, which can cause complications such as cardiovascular diseases, stroke, chronic kidney failure, foot ulcers and damage to the eyes (IDF, 2012). There are three main types of diabetes: type 1, in which the pancreas fails to produce enough insulin and which is generally genetically determined; type 2 diabetes is the seventh leading cause of death in the world (2.7% of the universal deaths, WHO, 2017), starts with insulin resistance (lack of insulin may also develop) and is promoted by obesity and a sedentary lifestyle (Coope et al., 2015); and gestational diabetes, an often transient disease that occurs when pregnant women develop a high blood sugar level (IDF, 2015; Folmer et al., 2017).

Floegel et al. (2012) investigated the association between the coffee consumption and the risk of chronic diseases, including type 2 diabetes. They used data from 42 659 participants collected over 8.9 years from the European Prospective Investigation into Cancer and Nutrition (EPIC) cohort. They found an inverse association of consumption of more than 4 cups (150 mL) of regular coffee per day with the overall risk of type 2 diabetes.

A number of similar studies have observed such effects related to regular coffee drinking. A recent meta-analysis of large epidemiological studies confirmed the link between moderate coffee consumption and a reduced risk of developing type 2 diabetes across different populations (Ding et al., 2014). The findings from these systematic studies demonstrate a clear inverse association between the coffee consumption and the risk of developing diabetes. Compared with no, or infrequent, coffee consumption, the risk of developing type 2 diabetes was reduced linearly, with a 33% reduction for 6 cups per day. In a similar comparison, drinking up to four cups per day of decaffeinated coffee was associated with a 20% reduced risk (Ding et al., 2014). This suggests that the protective effects of coffee on diabetes are independent of caffeine.

Animal studies have indicated that the main compounds responsible for the protective effect are chlorogenic acids (Kempf, 2010) and its derivatives, as well as trigonelline (van Dijk et al., 2009; Rios et al., 2015). They appear to preferentially target hepatic glucose metabolism by improving insulin sensitivity (Lecoultre et al., 2014). Other proposed mechanisms observed in *in vivo* and *in vitro* studies include the regulation of key enzymes of glucose and lipid metabolism, such as glucokinase, glucose-6-phosphatase, fatty acid synthase and carnitinepalmitoyltransferase (Waite, 2015). In a human study, trigonelline generated significantly lower glucose and insulin levels after an oral glucose load compared with a placebo (Rios et al., 2015).

3.5 Coffee and liver diseases

There are a number of diseases that can impact liver health and include both liver cancer and cirrhosis, a progressive disease caused by liver steatosis (fatty liver) and alcohol abuse, where the healthy tissue is replaced by the scar tissue and eventually prevents the liver from functioning correctly (Saab et al., 2014). According to a recent meta-analysis of 16 human studies, coffee consumption reduces the risk of developing liver cancer by 40% compared with no coffee consumption (Larsson and Wolk, 2007; Bravi et al., 2013).

In a clinical study performed in Brazil, caffeine consumption greater than 123 mg per day was also associated with reduced hepatic fibrosis (Machado et al., 2014). In addition, the study observed positive effects of regular coffee consumption in patients with chronic hepatitis C.

A number of *in vitro* studies have demonstrated the strong role of the chlorogenic acids present in coffee in protecting the liver from damage at various levels, possibly by preventing cell apoptosis and oxidative stress damage due to the activation of the body's natural antioxidant and anti-inflammatory systems (Ji et al., 2013). Coffee melanoidins have also been reported to have a protective effect on liver steatosis in obese rats (Vitaglione et al., 2012), which suggests that the melanoidins in coffee may have an influence on liver fat and functionality. Although melanoidins do not seem to be absorbed in humans, they can function as an antioxidant dietary fibre, like the unabsorbed portion of chlorogenic acids, quenching radicals and improving the reduced/oxidized glutathione balance in the colon. At the same time, they may act to promote the growth of a beneficial colon microbiota, affecting inflammatory pathways in the colon and consequently in the liver (Folmer et al., 2017).

3.6 Coffee and cancer

In the broadest sense, cancer represents the final result of abnormal cell growth and can occur in most human tissues. The carcinogenicity of coffee drinking was assessed by the IARC in 1991. At that time, coffee was classified as 'possibly carcinogenic to humans' (Group 2B), based on limited evidence of an association with cancer of the urinary bladder from case-control studies, and inadequate evidence of carcinogenicity in experimental animals. When subsequent studies were controlled for smoking, they failed to show an elevated risk of bladder cancer (Butt and Sultan, 2011). Recently, the IARC re-evaluated the carcinogenicity of drinking coffee and other hot beverages (Loomis et al., 2016), using a much larger database of more than 1000 observational and experimental studies. In assessing the accumulated epidemiological evidence, more weight was given to well-conducted prospective cohort and population-based case-control studies that controlled adequately for important potential confounding factors, including smoking (tobacco) and alcohol consumption. In conclusion, there was no consistent evidence associating drinking coffee with bladder cancer. In contrast, for endometrial cancer, the five largest cohort studies showed mostly inverse associations with coffee drinking. These results were supported by the findings of several case-control studies and a meta-analysis. Inverse associations with coffee drinking were also observed in cohort and case-control studies of liver cancer in Asia, Europe and North America. A meta-analysis of prospective cohort studies estimated that the risk of liver cancer decreases proportionally with coffee intake. No association or a modest inverse association for female breast cancers was found. Similarly, no association was found for pancreas and prostate cancers. Data were also available for more than 20 other cancers, including lung, colorectal, stomach, oesophageal, oral cavity, ovarian and brain cancers and childhood leukaemia. Although the volume of data for some of these cancers was substantial, evidence was inadequate for all the other cancers reviewed for reasons including inconsistency of findings across studies, inadequate control for potential confounding factors, potential for measurement error, selection bias or recall bias or insufficient numbers of studies (Loomis et al., 2016). As a result of this re-evaluation, coffee was upgraded by the IARC and is no longer considered to be potentially carcinogenic. In summary, epidemiological data demonstrated that coffee consumption is actually associated with a *lower* overall risk of cancer, especially liver and endometrial cancers.

There are several compounds in coffee that have been found to play a protective role against cancer, and the most well-known are chlorogenic acids and their derivatives. The contribution of melanoidins has also been suggested to decrease the risk of colon cancer (see chlorogenic acids and melanoidins topics in this chapter). Based on *in vitro* and animal evidence, coffee diterpenes are also strong candidates.

Epidemiologic studies have shown an increased risk of oesophageal cancer from drinking hot beverages such as maté, tea or coffee. It has been observed that the intra-oesophageal temperature is increased by 6–12°C when coffee was drunk at 65°C (Islami et al., 2009). The high-temperature injures the oesophageal mucosa and consequently causes inflammation or forms reactive nitrogen species, a type of free radical. It has been suggested by the IARC that drinking coffee, and other hot beverages, at temperatures above 65°C increases the risk of oesophageal cancer (Loomis et al., 2016). Although in some countries, coffee is consumed at temperatures below 65°C, in other countries, the temperature can be much higher. In public places, serving coffee at very high temperatures may influence people to drink it hotter than they would at home, so people should be aware of this factor for all hot beverages, soups and hot foods in general.

4 Potential side effects of coffee drinking

4.1 Hyper stimulation and sleep quality and duration by caffeine

Caffeinated coffee can cause irritability and anxiety, and reduce sleep quality by increasing the time required to fall asleep, interfering with the depth of sleep and reducing the total time spent sleeping. It can also cause more frequent awakening or sleep fragmentation (Folmer et al., 2017; Clarc and Landholt, 2016; Huang et al., 2011). The use of caffeine in energy drinks, and the risk of overdosing in children, motivated health authorities to evaluate and publish guidelines on safe caffeine consumption. The most recent report is the EFSA's 2015 scientific opinion on caffeine safety (EFSA 4102, 2015a). National health authorities have also published reports like the US Department of Agriculture report (2015). The general agreement is that the habitual consumption of up to 400 mg of caffeine per day, and up to 200 mg per serving, does not cause safety concerns for non-pregnant adults. Considering a range between 100 and 200 mg caffeine per cup, this would translate into 2–4 cups per day.

4.2 Caffeine tolerance, dependence and withdrawal

Caffeine is the most widely used psychoactive substance in the world, and the issue of possible dependence on caffeine has been discussed for many years. In fact, different drugs affect different people in different ways, and caffeine is no exception. It is therefore difficult to make general statements on dependence, tolerance and withdrawal; however, there is no such brain circuit that links caffeine to dependence. Caffeine does not affect areas involved in reinforcing and rewarding (Nehlig, 2010). According to the standard for measuring any potential drug abuse and dependence (as defined by the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV, American Psychiatric Association, 2000), there are no criteria that qualify caffeine for potential drug abuse (Folmer et al., 2017).

As with any drug, regular caffeine users will establish a partial tolerance to caffeine. However, studies have shown that this tolerance only applies to effects such as jitteriness, anxiety and an increased heart rate. Users do not develop a tolerance to the benefits of caffeine consumption such as improved mental performance, although sometimes slightly higher doses of caffeine are required (Satel, 2006).

The types of caffeine withdrawal symptoms which are most often reported are headaches; feelings of weariness, weakness and drowsiness; impaired concentration; fatigue and work difficulty; depression; anxiety; irritability; increased muscle tension and occasional tremors, nausea or vomiting. Withdrawal symptoms generally peak 20–48 h after the last caffeine was consumed, although users can generally avoid these if caffeine consumption is progressively decreased (Nehlig, 2010; Folmer et al., 2017).

Excessive coffee intake does not cause organic toxicity, but it can generate negative side effects, such as those associated with caffeine withdrawal. Symptoms related to the toxicity of coffee can occur at levels well below fatal doses; for example, concentrations above 15 mg caffeine per kg of bodyweight may be toxic for the cardiovascular, nervous and gastrointestinal systems (e.g. 1 g of caffeine for a person weighing 70 kg). Although such caffeine levels are not easily obtained through acute coffee intake, users may easily consume caffeine pills in such quantities. Reported overdose symptoms are hypertension or hypotension, tachycardia, vomiting, fever, delusion, hallucinations, arrhythmia, cardiac arrest, coma and death. Fatalities most commonly result from seizures and cardiac arrhythmias at plasma levels of 100–180 μg per mL (Childs and de Wit, 2012; Folmer et al., 2017). However, caffeine-related deaths have not been associated with coffee drinking (Yamamoto, 2015).

4.3 Cholesterol-raising effects of diterpenes

Epidemiologic and mechanistic studies have reported that the diterpenes, cafestol, and to a lesser extent, kahweol, naturally found in coffee oil and in unfiltered coffees, can alter lipid enzymes and thus influence cholesterol levels. This relationship was found to be linear with increasing cafestol consumption (Urgert and Katan, 1997). A meta-analysis of a set of 18 clinical intervention trials on coffee consumption and cholesterol and serum lipids was performed by Jee et al. (2001). The authors corroborated the dose–response relationship between coffee consumption and cholesterol and observed a strong increase upon the consumption of 6 or more cups of boiled coffee per day, which was not observed when a paper filter was used.

The high consumption of diterpenes has been associated with elevated homocysteine and low-density lipoprotein levels in human plasma, which may indirectly increase the risk of cardiovascular diseases (Farah, 2012).

4.4 Gastro-oesophageal reflux or heartburn

Gastro-oesophageal reflux, also called heartburn, is caused by the reflux of gastric fluid into the oesophagus due to the low pressure in the sphincter muscle at the junction of the stomach and oesophagus. A number of people suffering from this condition have mentioned that coffee may be one of the food products causing this complaint. Although some studies have tried to investigate this, the role of coffee consumption in reflux is still unclear. It has, for example, been reported that a few compounds in coffee stimulate the production of gastric juice, which is very acidic when first produced in the stomach (pH 1–2). Chlorogenic acids, Nβ-alkanoil-5-hydroxytryptamides (C5HTs) from coffee wax, and to a lesser extent, caffeine, are some of the main compounds thought to promote this effect (Fehlau and Netter, 1990). Additionally, it has been hypothesized that the roasting products of chlorogenic acid such as pyrogallol, like C5HTs, irritate the gastric mucosa (Darboven, 1997). In addition to the stimulation of gastric juice, coffee consumption also seems to cause muscle contraction impairment (relaxation effect) of the lower oesophageal sphincter in some individuals, promoting heartburn, which has been attributed to caffeine (Terry et al., 2000). The pH of a coffee brew is mildly acidic, commonly fluctuating between 5.8 and 5.5 in robusta coffees and between 4.3 and 4.8 in fresh lightly roasted acidic arabica coffees, with approximately pH 5.0 being more usual in dark roasted blends. This is much higher than the pH of gastric juice or, for example, the pH of apple juice (pH 4.3–3.3) or citric juice (pH 2.3–3.3) (Farah, unpublished).

Therefore, based on current knowledge, the stimulation of gastric juice production along with the relaxation of the oesophageal sphincter seems to be the most likely causes for heartburn, although some studies support the contribution of the acidity of foods to heartburn (Feldman and Barnett, 1995). Since there are only a few studies on this subject and none in humans, more mechanistic and clinical studies are necessary to prove the involvement of each of these specific coffee compounds in this disease, as well as improving conditions in their absence.

Some pre-roasting technological methods have been developed which aim to decrease heartburn, although no clinical studies have yet proved that these treatments are effective in humans. Reducing coffee wax, and thus C5HTs, can be achieved by applying steam treatment, either as a stand-alone process, or as part of the water or CO_2 decaffeination methodologies (which in addition decreases the content of chlorogenic acids and caffeine).

5 Final considerations

Since the initial studies published in medical journals in the eighteenth century, coffee has been through many waves of approval and disapproval. As science has evolved and confounding factors could be accounted for, an increasing number of studies have found correlations between coffee consumption and reduced risk of developing certain diseases. As *in vitro* and animal studies confirm the involvement of active coffee components in specific diseases, the challenge remains to fully understand the mechanisms that these active compounds exert, as coffee is a molecularly highly complex beverage made up of thousands of compounds. It is, however, becoming more evident that it is not only the specific compounds in coffee, but rather the beverage as a whole that is responsible for its beneficial effects.

Despite the potentially positive contribution of coffee to reducing the risk of certain diseases, these findings need to be related to each other and, more importantly, to lifestyle factors that influence the risk of developing certain diseases and longevity. Some of these include not smoking, good nutrition (a balanced and varied diet including five servings of fruit and vegetables daily), exercise, low alcohol consumption and low stress, all of which have a strong documented impact on disease prevention and life expectation (Khaw et al., 2008).

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7 Where to look for further information

7.1 Recommended books

Folmer, B. (Ed). (2017). *The Craft and Science of Coffee*. Elsevier, London,1st edition.
Farah, A. (Ed). (2017). *Coffee: Chemistry Quality and Health Implications*. Royal Society of Chemistry, UK, 1st edition, In press.
Preedy, V. (Ed). (2015). *Coffee in Health and Disease Prevention*. Elsevier, London, UK, 1st edition.
Feng, I. (2012). *Coffee: Emerging Health Effects and Disease Prevention*. IFT Press/Willey-Blackwell, USA.

7.2 Recommended websites

- International coffee organization (ICO): www.ico.org. The International Coffee Organization (ICO) is the main intergovernmental organization for coffee, bringing together producing and consuming countries to tackle the challenges faced by the world coffee sector through international cooperation. It makes a practical contribution to the world coffee economy and to improving the standards of living in developing countries by enabling government representatives to exchange views and coordinate coffee policies and priorities, and enabling government representatives to exchange views and coordinate coffee policies and priorities at regular high level. On this site, in addition to the information on world coffee production, exports and imports, general global information on coffee is also found.
- Association for Science and Information on Coffee (ASIC): www.asic-cafe.org. 'ASIC is a completely independent organization in the world whose scientific vocation is specifically devoted to the coffee tree, the coffee bean and the coffee drink'. On ASIC's site, you will find the proceedings of the previous colloquia as well as information and links on the latest publications on coffee agronomy, chemistry, technology, coffee and health and physiological effects of coffee.
- The Specialty Coffee Association (SCA) that is a membership-based association acts as a unifying force within the specialty coffee industry and works to make coffee better by raising standards worldwide through a collaborative and progressive approach. Members of the SCA include coffee retailers, roasters, producers, exporters and importers, as well as manufacturers of coffee equipment and related products. For more information, access www.sca.coffee.The SCA was formed in January 2017 following the merger of the SCAA and SCAE, acting in the United States and Europe. The respective websites are currently still active and information on training, education, events and standards can still be found (seewww.scaa.org and www.scae.org)
- A website fully dedicated to entire information on coffee and health is www.coffeeandhealth.org. It is a science-based resource developed for health care and other professional audiences and provides the latest information and research into coffee, caffeine and health.

8 References

ABIC, 2016 http ://abic.com.br/institucional/projetos-sociais/ accessed dez, 2016 and 2017.

Alcázar, A., Fernández-Cáceres, P. L., Martın-Valero, M., Pablos, F. and González, A. G., 2003. Ion chromatographic determination of some organic acids, chloride and phosphate in coffee and tea. *Talanta, Amsterdam* 61(1), 95–101.

American Psychiatric Association. 2000. *Diagnostic and Statistical Manual of Mental Disorders*, 4th Ed. American Psychiatric Association, Arlington, VA.

Antonio, A. G., Moraes, R. S., Perrone, D., Maia, L. C., Santos, K. R. N., Iório, N. L. P. and Farah, A., 2010. Species, roasting degree and decaffeination influence the antibacterial activity of coffee against Streptococcus mutans. *Food Chemistry* 118, 782–8.

Andersen, L. F., Jacobs Jr., D. R., Carlsen, M. H. and Blomhoff, R., 2006. Consumption of coffee is associated with reduced risk of death attributed to inflammatory and cardiovascular diseases in the Iowa Women's Health Study. *American Journal of ClinicalNutrition* 83(5), 1039–46.

Arab, L., Khan, F' and Lam, H., 2013. Epidemiologic evidence of a relationship between tea, coffee, or caffeine consumption and cognitive decline. *Advances in Nutrition* 4(1), 115–22.

Arauz, J., Rivera-Espinoza, Y., Shibayama, M. and Muriel, P., 2015. Nicotinic acid prevents experimental liver fibrosis by attenuating the prooxidant process. *International Immunopharmacology* 28(1), 244–51.

Arendash, G. W. and Cao, C., 2010. Caffeine and coffee as therapeutics against Alzheimer's disease. *Journal of Alzheimer's Disease* 20(S1), 117–26.

Balzer, H. H., 2001. Acids in coffee. In: *Coffee Recent Developments* (Clarke, R. J. and Vitzthum, O. G. (Eds). Blackwell Science, Berlin, p. 18.

Barranco Quintana, J. L., Allam, M. F., SerranoDel Castillo, A. and Fernandez-Crehuet Navajas, R., 2007. Alzheimer's disease and coffee: A quantitative review. *Neurological Research* 29, 91–5.

Bizzo, M. L. G., Farah, A., Kemp, J. A. and Scancetti, L. B., 2015. Highlight in the history of coffee science related to health. In: *Coffee in Health and Disease Prevention* (Preedy, V. (Ed.)). Elsevier, London, UK, pp.11–18.

Borrelli, R. C., Esposito, F., Napolitano, A., Ritieni, A. and Fogliano, V., 2004. Characterization of a new potential functional ingredient: Coffee silverskin. *Journal of Agriculture and Food Chemistry* 52, 1338–43.

Boekshoten, M.V., Van Cruschten, S.T., Kosmeijer-Schuil, T.G. and Katan, M.B. 2006. Negligible amounts of cholesterol-raising diterpenes in coffee made with coffee pads in comparison with unfiltered coffee, 2006. *Nederlands Tijdschrift voor Geneeskunde*, 150, 2873–5.

Borota, D., Murray, E., Kecell, G., Chang, A., Watabe, J. M., Ly, M., Toscano, J. P. and Yassa, M.A., 2014. Post-study caffeine administration enhances memory consolidation in humans. *Nature Neuroscience* 17, 201–3.

Bravi, F., Bosetti, C., Tavoni, A., Gallus, S. and La Vecchia, C., 2013.Coffee reduces risk for hepatocellular carcinoma: An updated meta-analysis. *Clinical Gastroenterology and Hepatology* 11, 1413–21.

Butt, M. S. and Sultan, M. T., 2011. Coffee and it's Consumption: Benefits and Risks. *Critical Reviews in Food Science and Nutrition* 51, 363–73.

Camfield, D. A., Silber, B. Y., Scholey, A. B., Nolidin, K., Goh, A. and Stough, C., 2013. A Randomised placebo-controlled trial to differentiate the acute cognitive and mood effects of chlorogenic acid from decaffeinated coffee. *Public Library of Science One* 8(12), e82897.

Carpenter, K. J., 1983. The relationship of pellagra to corn and the low availability of niacin in cereals. *Experientia Suppl*. 44: 197–222.

Casal S., 2017. Potential effects of β-carbolines on human health. In *Coffee: Chemistry, Quality, and Health Implications* (Farah, A. (Ed.)). Royal Society of Chemistry, London, UK. In press.

Chang, W. H., Hu, S. P., Huang, Y. F., Yeh, T. S. and Liu, J. F., 2010. Effect of purple sweet potato leaves consumption on exercise-induced oxidative stress and IL-6 and HSP72 levels. *Journal of Applied Physiology* 109(6), 1710–15.

Childs, E. and de Wit, H., 2012. Potential mental risks. In: *Coffee, Emerging Health Effects and Disease Prevention* (Chu, Y.-F. (Ed.)). Wiley-Blackwell, Oxford, UK, pp. 293–306.

Clarc, I. and Landholt, H. P., 2017. Coffee, caffeine and sleep: A systematic review of epidemiological studies and randomized controlled trials. *Sleep Medicine Reviews* 31, 70–8.
Cliford, M. N. 1985. Chlorogenic acids in coffee. In *Chemistry*, vol 1, Clarke R. J. and Macrae R. (Eds), Elsevier Applied Science, London, 153–202.
Coope, A., Torsoni, A. S. and Velloso, L., 2015. Mechanisms in endocrinology: Metabolic and inflammatory pathways on the pathogenesis of type 2 diabetes. *European Journal of Endocrinology* 15, 1065.
Corley, J., Jia, X., Kyle, J. A., Gow, A. J., Brett, C. E., Starr, J. M., McNeill, G. and Deary, I. J. 2010. Caffeine consumption and cognitive function at age 70: The Lothian Birth Cohort 1936 study. *Psychosomatic Medicine* 72, 206–14.
Costa, J., Lunet, N., Santos, C., Santos, J. and Vaz-Cameiro, A., 2010. Caffeine exposure and the risk of Parkinson's disease: A systemic review and meta-analysis of observational studies. *Journal of Alzheimer's Disease* 20, S221–38.
Costill, D. L., Dalsky, G. P. and Fink, W. J., 1978. Effects of caffeine ingestion on metabolism and exercise performance. Medicine *Science in Sports* 10(3), 155–8.
Crippa, A., Discacciati, A., Larsson, S. C., Wolk, A. and Orsini, N., 2014. Coffee consumption and mortality from all causes, cardiovascular disease, and cancer: A dose-response meta-analysis. *American Journal of Epidemiology* 180, 763–75.
Cropley, V., Croft, R., Silber, B., Neale, C., Scholey, A., Stough, C. and Schmitt, J., 2012. Does coffee enriched with chlorogenic acids improve mood and cognition after acute administration in healthy elderly? A pilot study. *Psychopharmacology* 219(3), 737–49.
Cunha, S. and Fernandes, J. 2017. Pesticides. In: *Coffee: Chemistry, Quality and Health Implications* (Ed.: Farah, A.). Royal Society of Chemistry, London, UK. In press.
Darboven, A.1997. Method for the quality improvement of raw coffee by treatment with steam and water (in German). Europaisches Patent- € blatt, 1997/05/95109295.6.
Ding, M., Satija, A., Bhupathiraju, S. N., Hu, Y., Sun, Q., Han, J., Lopez-Garcia, E., Willet, W., van Dam. R. M. and Hu, F. A., 2015. Association of coffee consumption with total and cause-specific mortality in three large prospective cohorts.*Circulation* 132(24), 2305–15.
Ding, M., Bhupathiraju, S.N., Chen, M., vanDam, R. M. and Hu, F. B., 2014. Caffeinated and decaffeinated coffee consumption and risk of Type 2 diabetes: A systematic review and a dose-response meta-analysis. *Diabetes Care* 37(2), 569–86.
Drapeau, C., Hamel-Hebert, I., Robillard, R., Seimaoui, B., Filipini, D. and Carner, J., 2006. Challenging sleep in aging: the effects of 200 mg of caffeine during the evening in young and middle-aged moderate caffeine consumers. *Journal of Sleep Research* 15(2), 133–41.
Einother, S. J. L. and Giesbrecht, T., 2012. Caffeine as an attention enhancer: Reviewing existing assumptions. *Psychopharmacology* 225(2), 251–74.
Esposito, E., Rotilio, D., Di Matteo, V., Di Giulio, C., Cacchio, M. and Algeri, S., 2002. A review of specific dietary antioxidants and the effects on biochemical mechanisms related to neurodegenerative processes. *Neurobiology of Aging* 23, 719–35.
EFSA (European Food Safety Authority), 2008. Scientific opinion of the panel on contaminants in the food chain on a request from the European commission on polycyclic aromatic hydrocarbons in food. *The EFSA Journal*, 724, 1–114.
EFSA (European Food Safety Authority), 2011a. Scientific Opinion on Flavouring Group Evaluation 218, Revision 1 (FGE.218Rev1): alpha, beta-Unsaturated aldehydes and precursors from subgroup 4.2 of FGE.19: Furfural derivatives. *EFSA Journal* 9(3), 1840.
EFSA (European Food Safety Authority), 2011b. EFSA Panel on Dietetic Products, Nutrition and Allergies (NDA), Scientific Opinion on the substantiation of health claims related to caffeine and increase in physical performance during short-term high-intensity exercise (ID 737, 1486, 1489), increase in endurance performance (ID 737, 1486), increase in endurance capacity (ID 1488) and reduction in the rated perceived exertion/effort during exercise (ID 1488, 1490) pursuant to Article 13(1) of Regulation (EC) No 1924/20061. *EFSA Journal* 9(4), 2053.
EFSA (European Food Safety Authority), 2011c. EFSA Panel on Dietetic Products, Nutrition and Allergies (NDA), Scientific Opinion on the substantiation of health claims related to caffeine and increased fat oxidation leading to a reduction in body fat mass (ID 735, 1484), increased energy

expenditure leading to a reduction in body weight (ID 1487), increased alertness (ID 736, 1101, 1187, 1485, 1491, 2063, 2103) and increased attention (ID 736, 1485, 1491, 2375) pursuant to Article 13 (1) of Regulation (EC) No 1924/20061. *EFSA Journal* 9(4), 2054.

EFSA (European Food Safety Authority), 2015a. EFSA panel on dietetic products, Nutrition and Allergies (NDA), scientific opinion on the safety of caffeine. *EFSA Journal* 13(5), 4102.

EFSA (European Food Safety Authority), 2015b. EFSA panel on contaminants in the food chain (CONTAM) scientific opinion on acrylamides in food. *EFSA Journal* 13(6), 4104.

FDA (Food and Drug Administration), 2016. *FDA Guidance for Industry Acrylamide in Foods*. FDA Office of Food Safety, USA.

Farah, A. and Donangelo, C., 2006. Phenolic compounds in coffee. *Brazilian Journal of Plant Physiology* 18(1), 23–36.

Farah, A., 2012. Coffee constituents. In: Chu, Y.-F. (Ed.), *Coffee: Emerging Health Effects and Disease Prevention*. IFT Press/Willey-Blackwell, USA, pp. 21–58.

Fehlau, R. and Netter, K. J.1990. Effect of untreated and non-irritating purified coffee and carbonic acid hydrytryptamides on the gastric mucosa in the rat Z. *Gastroenterol*, 28, 234–8.

Feldman, M. and Barnett, C., 1995. Relationships between the acidity and osmolarity of popular beverages and reported postprandial heartburn. *Gastroenterology*, 108, 125–31.

Feng, R., Lu, Y., Bowman, L. L., Qian, Y., Castranova, V. and Ding, M., 2005. Inhibition of activator Protein-1, NF-κB, and MAPKs and induction of phase 2 detoxifying enzyme activity by chlorogenic acid. *Journal of Biological Chemistry* 280, 27888–95.

Ferreira, I. M. P. L. V. O., Pinho, O. and Petisca, C., 2017. Potential effects of furan and related compounds on health. In: *Coffee: Chemistry, Quality and Health Implications* (Ed.: Farah, A.). Royal Society of Chemistry, London, UK. In press.

Floegel, A., Pischon, T., Bergmann, M. M., Teucher, B., Kaaks, R, Boeing, H., 2012. Coffee consumption and risk of chronic disease in the European Prospective Investigation into Cancer and Nutrition (EPIC)–Germany study. *American Journal Clinical Nutrition* 95, 901–8.

Fogliano, V. and Morales, F. J., 2011. Estimation of dietary intake of melanoidins from coffee and bread. *Food and Function* 2, 117–23.

Folmer, B., Farah, A., Fogliano, V. and Jones, L. 2017. Human wellbeing – Sociability, performance and health. In: *The Craft and Science of Coffee*, 1st Ed. (Ed.: Folmer, B.). Elsevier, London, UK, pp. 493–510.

Freedman, N., Park, Y., Abnet, C. C., Hollenbeck, A. R. and Sinha, R., 2012. Association of coffee drinking with total and cause-specific mortality. *New England Journal of Medicine* 366(20), 1891–904.

Garfinkel, B. D., Webster, C. D. and Sloman, L., 1981. Responses to methylphenidate and various does of caffeine in children with attention deficit disorder. *The Canadian Journal of Psychiatry/ La Revue canadienne de psychiatrie* 26(6), 395–401.

Garsetti, M., Pellegrini, N., Baggio, C. and Brighenti, F., 2000. Antioxidant activity in human faeces. *British Journal of Nutrition* 84(5), 705–10.

Gebicki, J., Marcinek, A., Chlopicki S. and Adamus, J. 2008. The use of quaternary pyridinium compounds for vasoprotection and/or hepatoprotection. Patent WO2008104920A1.

Gerson, M., 1978. The cure of advanced cancer by diet therapy: a summary for 30 years of clinical experimentation. *Physiological Chemistry and Physics* 10, 449–64.

Glória and Engeseth , 2017a. Potential beneficial effects of bioactive amines on health. In:*Coffee: Chemistry, Quality, and Health Implications* (Ed.: Farah, A.). Royal Society of Chemistry, London, UK. In press.

Glória and Engeseth , 2017b. Potential adverse effects of coffee bioactive amines to human health. In *Coffee: Chemistry, Quality, and Health Implications* (Ed.: Farah, A.). Royal Society of Chemistry, London, UK. In press.

Gniechwitz, D., Brueckel, B., Reichardt, N., Blaut, M., Steinhart, H. and Bunzel, M. 2007. Coffee dietary fiber contents and structural characteristics as influenced by coffee type and technological and brewing procedures. *Journal of Agricultural and Food Chemistry* 55, 11027–34.

Gniechwitz, D., Reichardt, N., Ralph, J., Blaut, M., Steinhart, H.and Bunzel, M., 2008. Isolation and characterisation of a coffee melanoidin fraction. *Journal of the Science of Food Agriculture* 88, 2153–60.

Goldstein, E. R., Ziegenfuss, T., Kalman, D., Kreider, R., Campbell, B., Wilborn, C., Taylor, L., Willoughby, D., Stout, J., Graves, B. S., Wildman, R., Ivy, J. L., Spano, M., Smith, S. E. and Antonio, J., 2010. International society of sports nutrition position stand: Caffeine and performance. *Journal of the International Society of Sports Nutrition* 7, 5.

Grembecka, M., Malinowska, E. and Szefer, P., 2007. Differentiation of market coffee and its infusions in view of their mineral composition. *Science of the Total Environment* 383, 59–69.

Gross, G., Jaccaud, E. and Hugget, A.C., 1997. Analysis of the content of the diterpenes cafestol and kahweol in coffee brews. *Food and Chemical Toxicology* 35, 547–54.

Health Canada, 2011. Information for Parents on Caffeine in Energy Drinks.www.hc-sc.gc.ca/fn-an/securit/addit/caf/faq-eng.php, Accessed January 2017.

Hedden, T. and Gabrieli, J. D. E., 2004. Insights into the ageing mind: A view from cognitive neuroscience. *Nature Reviews Neuroscience* 5, 87–97.

Higdon, J. V. and Frei, B., 2006. Coffee and health: A review of recent human research. *Critical Reviews in Food Science and Nutrition* 46, 101–23.

Hirakawa, N., Okauchi, R., Miura, Y. and Yagasaki, K., 2005. Anti-invasive activity of niacin and trigonelline against cancer cells. *Bioscience, Biotechnology and Biochemistry* 69(3), 653–8.

Hogervost, E., Bandelow, S., Schmitt, J., Jentjens, R., Oliveira, M., Allgrove, J., Carter, T. and Gleeson, M., 2008. Caffeine improves physical and cognitive performance during exhaustive exercise. *Medicine and Science in Sports and Exercise* 40, 1841–51.

Hong, B. N., Yi, T. H., Park, R., Kim, S. Y. and Kang, T. H., 2008. Coffee improves auditory neuropathy in diabetic mice. *Neuroscience Letters* 441(3), 302–6.

Huang, Z., Urade, Y. and Hayaishi, O., 2011. The role of adenosine in the regulation of sleep. *Current Topics in Medicinal Chemistry* 11, 1047–57.

International Agency for Research on Cancer (IARC), 2010. Some non-heterocyclic polycyclic aromatic hydrocarbons and some related exposures. IARC Monogr Eval Carcinog Ris 1997; USDA National Nutrient Database for Standard Reference, release 28, 2015 any Maryland, USA

International Coffee Organization, 2014. World coffee trade (1923–2013): A review of the markets, challenges and opportunities facing the sector. 112thSession.http://www.ico.org/news/icc-111-5-r1e-world-coffee-outlook.pdf

International Diabetes Federation (IDF), 2012. *Clinical Guidelines Task Force Global Guideline for Type 2 Diabetes*.

International Diabetes Federation (IDF), 2015. *Diabetes Atlas*, 5th Ed. Brussels, Belgium.

Islami, F., Boffetta, P., Ren, J. S., Pedoeim, L., Khatib, D. and Kamangar, F., 2009. High-temperature beverages and foods and esophageal cancer risk – A systematic review. *International Journal of Cancer* 125(3), 491–524.

Jee, S. H., He, J., Appel, L. J., Whelton, P. K., Suh, I. and Klag, M. J., 2001. Coffee consumption and serum lipids: A meta-analysis of randomized controlled clinical trials. *American Journal of Epidemiology* 153, 353–62.

Jenkins, N. T., Trilk, J. L., Singhal, A., O'Connor, P. J. and Cureton, K. J., 2008. Ergogenic effects of low doses of caffeine on cycling performance. *International Journal of Sport Nutrition and Exercise Metabolism* 18(3), 328–42.

Ji L., Jiang, P., Lu, B., Sheng, Y., Wang, X. and Wang, Z., 2013. Chlorogenic acid, a dietary polyphenol, protects acetaminophen-induced liver injury and its mechanism. *Journal of Nutrition and Biochemistry* 24(11), 1911–19.

Jurkowska, R. Z., Jurkowski, T. P. and Jeltsch, A., 2011. Structure and function of mammalian DNA methyltransferases. *European Journal of Chemical Biology* 12, 206–22.

Kalaska, B., Piotrowski, L., Leszczynska, A., Michalowsk, B., Kramkowski, K., Kaminski, T., Adamus, J., Marcinek A., Gebicki, J., Mogielnicki A. and Buczko, W., 2014. Antithrombotic effects of pyridinium compounds formed from trigonelline upon coffee roasting. *Journal of Agricultural and Food Chemistry*, 62(13), 2853–60.

Kasai H., Fukada, S., Yamaizumi, Z., Sugie, S. and Mori, H. 2000. Action of chlorogenic acid in vegetables and fruits as an inhibitor of 8-hydroxydeoxyguanosine formation in vitro and in a rat carcinogenesis model. *Food and Chemical Toxicology* 38, 467–71.

Kempf, K., Herder, C., Erlund, I., Kolb, H., Martin, S., Carstensen, M., Koenig, W., Sandwall, J., Bidel, S., Kuha, S. and Tuomilehto, J., 2010. Effects of coffee consumption on subclinical inflammation and other risk factors for type 2 diabetes: A clinical trial. *American Journal of Clinical Nutrition* 91, 950–7.

Khaw, K. T., Wareham, N., Bingham, S., Welch, A., Luben, R. and Day, N., 2008. Combined impact of health behaviours and mortality in men and women: the EPIC-Norfolk prospective population study. *PLoS Medicine* 5(1), e12.

Kuczkowski, K. M., 2009. Caffeine in pregnancy.*Archives of Gynecology and Obstetrics* 280, 695–8.

Kuwana, Y., Shiozaki, S., Kanda, T., Kurokawa, M., Koga, K., Ochi, M., Ikeda, K., Kase, H., Jackson, M. J., Smith, L. A., Pearce, R. K. and Jenner, P. G.1999. Antiparkinsonian activity of adenosine A2A antagonists in experimental models. *Advances in Neurology*, 80, 121–3.

Lachenmeier, D.W., 2015. Furan in coffee products: A probabilistic exposure estimation. In:*Coffee in Health andDiseasePrevention* (Ed.:Preedy, V.). Elsevier, 1st edition. New York and London, pp. 887–93.

Lakey-Beitia, J., Berrocal, R., Rao, K. S.and Durant, A. A., 2015. Polyphenols as therapeutic molecules in Alzheimer's disease through modulating amyloid pathways. *Molecular Neurobiology* 51(2), 466–79.

Lang, R., Wahl, A., Stark, T. and Hofmann, T.2011. Urinary N-methylpyridinium and trigonelline as candidate dietary biomarkers of coffee consumption. *Molecular Nutrition & Food Research* 5(11), 1613–23.

Lang, R., Bardelmeier, I., Weiss, C., Rubach, M., Somoza, V. and Hofmann, T., 2010. Quantitation of $^{\beta}$N-Alkanoyl-5-hydroxytryptamides in coffee by means of LC-MS/MS-SIDA and assessment of their gastric acid secretion potential using the HGT-1 cell assay. *Journal of Agricultural and Food Chemistry* 58(3), 1593–602.

Larsson, S. C. and Wolk, A., 2007. Coffee consumption and risk of liver cancer: A meta-analysis. *Gastroenterology* 132, 1740–5.

Larsson S. C. and Orsini, N., 2011. Coffee consumption and risk of stroke: a dose-response meta-analysis of prospective studies. *American Journal of Epidemiology* 174(9), 993–1001

Le Bloch J. L. V., Chetiveaux M., Freuchet B., Magot T., Krempf M., Nguyen P. and Ouguerram K. 2010. Nicotinic acid decreases apolipoprotein B100-containing lipoprotein levels by reducing hepatic very low density lipoprotein secretion through a possible diacylglycerol acyltransferase 2 inhibition in obese dogs. *The journal of Pharmacology and Experimental Therapeutics*, 334, 583–89.

Lecoultre, V., Carrel, G., Egli, L., Binnert, C., Boss, A., MacMillan, E. L., Kreis, R., Boesch, C., Darimont, C. and Tappy, L., 2014. Coffee consumption attenuates short-term fructose-induced liver insulin resistance in healthy men. *American Journal of Clinical Nutrition* 99(2), 268–75.

Lipworth, L., Sonderman, J. S., Tarone, R. E. and McLaughlin, J., 2012. Review of epidemiologic studies of dietary acrylamide intake and the risk of cancer. *European Journal of Cancer Prevention* 21, 375–86.

Loomis, D., Kathryn, Z., Grosse, G. Y., Lauby-Secretan, B., El Ghissassi, F., Bouvard, V.,et al. 2016. Carcinogenicity of drinking coffee, mate, and very hot beverages.*The Lancet Oncology* 17(7),877–8.

Lopez-Garcia, E., Rodriguez-Artalejo, F., Rexrode, K. M., Logroscino, G., Hu, F. B. and van Dam, R. M., 2009. Coffee consumption and risk of stroke in women. *Circulation* 119(8),1116–23.

Ludwig, I. A., Clifford, M. N., Lean, M. E. J., Ashihara, H. and Crozier, A., 2014. Coffee: Biochemistry and potential impact on health.*Food and Function* 5, 1695–717.

Machado, S. R., Parise, E. R. and Carvalho, L., 2014. Coffee has hepatoprotective benefits in Brazilian patients with chronic hepatitis C even in lower daily consumption than in American and European populations. *Brazilian Journal of Infectious Disease* 18(2), 170–6.

Macrae, R., 1985. Nitrogenous compounds. In: *Coffee. Volume 1 Chemistry*, 1st Ed. (Ed.:Clarke, R. J.and Macrae, R.). Elsevier, London and New York, p.115.
Malerba, S., Turati, F., Galeone, C., Pelucchi, C., Verga, F., La Vecchia, C. and Tavani, A., 2013. A meta-analysis of prospective studies of coffee consumption and mortality for all causes, cancers and cardiovascular diseases.*European Journal of Epidemiology* 28(7), 527–39.
Moreira, A., Coimbra, M., Nunes, F. M., Passos, C. P., Santos, S. A., Silvestre, A. J., Silva, A., Rangel, M.and Domingues, M. R. M., 2015. Chlorogenic acid-arabinose hybrid domains in coffee melanoidins: Evidences from a model system. *Food Chemistry* 185, 135–44.
Moura-Nunes, N., Perrone, D., Farah, A. and Donangelo, C., 2009. The increase in human plasma antioxidant capacity after acute coffee intake is not associated with endogenous non-enzymatic antioxidant components. *International Journal of Food Sciences and Nutrition* 60(supp 6), 173–81.
Mucci, L. A. and Adami, H. O., 2009. The Plight of the Potato: Is dietary acrylamide a risk factor for human cancer?*Journal of National Cancer Institute* 101(9), 618–21.
Navarini, L., Colomban, S., Caprioli, G.and Sagratini, G., 2017. Isoflavones, Lignans and other minor polyphenols. In:*Coffee: Chemistry, Quality and Health Implications* (Ed. Farah, A.). Royal Society of Chemistry, London, UK. In press.
Nathan, J., Panjwani, S., Mohan, V., Joshi, V. and Thakurdesai, P. A., 2014. Efficacy and safety of standardized extract of Trigonella foenum-graecum L seeds in an adjuvant to L-Dopa in the management of patients with Parkinson's disease. *Phytotherapy Research* 28(2), 172–8.
Nawrot, P., Jordan, S., Eastwood, J., Rotstein, J, Hugenholtz, A. and Feeley, M., 2003. Effects of caffeine on human health. *Food Additives and Contaminants* 20, 1–30.
Nehlig A. and Debry, G., 1994. Consequences on the newborn of chronic maternal consumption of coffee during gestation and lactation: A review. *Journal of the American College of Nutrition* 13, 6–21.
Nehlig, A., 2010. Is caffeine a cognitive enhancer? *Journal of Alzheimer's Disease* 20(S1), 85–94.
Nkondjock, A., 2012. Coffee and cancers. In:*Coffee: Emerging Health Effects and Disease Prevention* (Ed.: Chu, Y.-F.). 1st edition, Wiley-Blackwell, Oxford, UK, pp. 293–306.
Nunes F. M., Coimbra M. A., Duarte A. C., and Delgadillo I., 1997. Foamability, foam stability, and chemical composition of espresso coffee as affected by the degree of roast, *J. Agric. Food Chem.*, 45 (8), 3238–43.
Oboh, G., Agunloye, O. M., Akinyemi, A. J., Ademiluyi, A. O. and Adefegha, S. A., 2013. Comparative study on the inhibitory effect of caffeic and chlorogenic acids on key enzymes linked to Alzheimer's disease and some pro-oxidant induced oxidative stress in rats' brain-in vitro. *Neurochemistry Research* 38(2), 413–19.
Obuleso, M., Dowlathabad, M. R. and Bramhachari, P. V., 2011. Carotenoids and Alzheimer's Disease: An insight into therapeutic role of retinoids in animal models. *Neurochemistry International* 59(5), 535–41.
O'Connor, P. J., Motl, R. W., Broglio, S. P. and Ely, M. R., 2004. Dose-dependent effect of caffeine on reducing leg muscle pain during cycling exercise is unrelated to systolic blood pressure. *Pain* 109, 291–8.
Oliveira, M., Casal, S., Morais, S., Alves, C., Dias, F., Ramos, S., Mendes, E., Delerue-Matos, C. and Oliveira, B. P. P., 2012. Intra- and interspecific mineral composition variability of commercial coffees and coffee substitutes. Contribution to mineral intake. *Food Chemistry* 130, 702–9.
Oliveira, M., Ramos, S., Delerue-Matos, C. and Morais S, 2015. Espresso beverages of pure origin coffee: Mineral characterization, contribution for mineral intake and geographical discrimination. *Food Chemistry* 177, 330–8.
Olson, C. A., Thornton, J. A., Adam, G. E. and Lieberman, H. R., 2010. Effects of 2 adenosine antagonists, quercetin and cafeïne, on vigilance and mood. *Journal of Clinical Psychopharmacology* 30(5), 573–8.
Passos, C. P., Cepeda, M. R., Ferreira, S. S., Nunes, F. M., Evtuguin, D. V., Madureira, P., Vilanova, M.and Coimbra, M. A., 2014. Influence of molecular weight on in vitro immunostimulatory properties of instant coffee. *Food Chemistry* 161, 60–6.

Petraco, M., 2005. The cup. In: *Espresso Coffee: The Science of Quality*, 2nd edition (Eds: Illy, A.and Viani, R.). Elsevier Academic Press, Italy, pp. 290–8.

Quan H.Y., Kim D.Y., and Chung S.H, 2013. Caffeine attenuates lipid accumulation via activation of AMP-activated protein kinase signaling pathway in HepG2 cells *BMB Reports* 46(4), 207–12.

Ramassamy, C., 2006. Emerging role of polyphenolic compounds in the treatment of neurodegenerative diseases: A review of their intracellular targets. *European Journal of Pharmacology* 545, 51–64.

Ramos, S., 2008. Cancer chemoprevention and chemotherapy: Dietary polyphenols and signaling pathways. *Molecular Nutrition and Food Research*, 52, 507–26.

Rebello, S.A. and Van Dam, R., 2013. Coffee consumption and cardiovascular health: Getting to the heart of the matter. *Current Cardiology Reports* 15(10), 403–5.

Riedel, A., Hochkogler C. M., Lang, R., Bytof, G., Lantz, I., Hofmann, T. and Somoza, V., 2014. N-methylpyridinium, a degradation product of trigonelline upon coffee roasting, stimulates respiratory activity and promotes glucose utilization in HepG2 cells. *Food &Function* 5(3):454–62.

Rios, J. L., Francini, F. and Schinella, G. R., 2015. Natural products for the treatment of Type 2 diabete mellitus. *Planta Medica* 81(12–13), 975–94.

Ritchie, K., Carriere, I., de Mendonca, A., Portet, F., Dartigues, J. F., Rouaud, O., Barberger-Gateau, P. and Ancelin, M. L., 2007. The neuroprotective effects of caffeine. A prospective population study. *Neurology* 69, 536–45.

Rodrigues, D.A.C.and Casal, S., 2017. β-Carbolines. In: *Coffee, Chemistry, Quality and Health Implications* (Ed.: Farah, A.). Royal Society of Chemistry, London, UK. In Press.

Rosso, A., Mossey, J. and Lippa, C. F., 2008. Review: Caffeine: Neuroprotective functions in cognition and alzheimer's disease. *American Journal of Alzheimer's Disease and Other Dementias* 23(5), 417–22.

Rubach M., Lang R., Bytof G., Stiebitz H., Lantz I., Hofmann T. and Somoza V. 2014. A dark brown roast coffee blend is less effective at stimulating gastric acid secretion in healthy volunteers compared to a medium roast market blend. *Mol Nutr Food Res.*, 58(6), 1370–3.

Rufián-Henares, J.A. and Morales, F., 2007. Functional Properties of melanoidins: In vitro antioxidant, antimicrobial and antihypertensive activities. *Food Research International*, 40(8), 995–1002.

Ryan, L., Hatfield, C. and Hofstetter, M., 2002. Caffeine reduces time-of-day effects on memory performance in older adults. *Psychological Science*, 13(1), 68–71.

Saab, S., Mallam, D., Cox, G.A. and Tong, M.J., 2014. Impact of coffee on liver diseases: A systematic review. *Liver International*, 34(4), 495–504.

Sales, A., Miguel, M. A. and Farah, A., 2017. Potential prebiotic effect of coffee. In *Coffee: Chemistry Quality and Health Implications* (Ed.: Farah, A.). Royal Society of Chemistry, London, UK. In press.

Santos, M. D., Almeida, M. C., Lopes, N. P. and Souza, G. E. P. 2006, Evaluation of the anti-inflammatory, analgesic and antipyretic activities of the natural polyphenol chlorogenic acid. *Biological and Pharmaceutical Bulletin* 29, 2236–40.

Santos, C., Costa, J., Santos, J., Vaz-Carneiro, A. and Lunet, N., 2010. Caffeine intake and dementia: Systematic review and meta-analysis. *Journal of Alzheimer's Disease* 20(S1), 187–204.

Santos, I. S., Matijasevich, A. and Domingues, M.R., 2012. Maternal caffeine consumption and infant nighttime waking: Prospective cohort study. *Pediatrics* 129, 860–8.

Satel, S., 2006. Is caffeine addictive? – a review of the literature. *American Journal of Drug and Alcohol Abuse* 32 (4), 493–502.

Sengpiel. V., Elind, E., Bacelis, J., Nilsson, S., Grove, J., Myhre, R., Haugen, M., Meltzer, H. M., Alexander, T., Jacobsson, B. and Brantsaeter, A. L., 2013. Maternal caffeine intake during pregnancy is associated with birth weight but not with gestational length: results from a large prospective observational cohort study. *BioMed Central Medicine* 11, 42–60.

Smith, A., Sutherland, D. and Christopher, G., 2005. Effects of repeated doses of caffeïne and mood and performance of alert and fatigued volunteers. *Journal of Psychopharmacology* 19(6), 620–6.

Smith, R. F., 1987. A History of coffee. In:*Coffee: Botany, Biochemistry and Production of Beans and Beverage* (Clifford, M. N.and Wilson, K. C. (Eds)). 1st edition, Croom Helm, Ney York, pp. 1–12.

Somoza V., Lindenmeier M., Wenzel E., Frank O., Erbersdobler H. F., and Hofmann T. (2003). Activity-guided identification of a chemopreventive compound in coffee beverage using in vitro and in vivo techniques. *J. Agric. Food Chem.*, 51 (23), 6861–9.

Speer, K. and Kölling-Speer, I.2017. Lipids. In: *Coffee: Chemistry, Quality, and Health Implications* (Ed.: Farah, A.).Royal Society of Chemistry, London, UK. In press.

Spriet, L. L. and Gibala, M. J., 2004. Nutritional strategies to influence adaptations to training.*Journal of Sports Sciences* 22, 127–41.

Stavchansky, S., Combs, A., Sagraves, R., Delgado, M. and Joshi, A., 1988. Pharmacokinetics of caffeine in breast milk and plasma after single oral administration of caffeine to lactating mothers. *Biopharmaceutics and Drug Disposition* 9, 285–99.

Terry, P., Lagergren, J., Wolk, A. and Nyren, O., 2000. Reflux-inducing dietary factors and risk of adenocarcinoma of the esophagus and gastric cardia. *Nutrition and Cancer* 38(2), 186–91.

Tohda, C., Kuboyama, T. and Komatsu, K., 2005. Search for natural products related to regeneration of the neuronal network. *Neurosignals* 14(1–2), 34–45.

Torres, T. and Farah, A., 2016. Coffee, maté, açaí and beans are the main contributors to the antioxidant capacity of Brazilian's diet. *European Journal of Nutrition* March 14. 56(4):1523–33. doi:10.1007/s00394-016-1198-9.

Trugo, L. C., 1985. Carbohydrates. In: *Coffee. Volume 1 Chemistry*, 1st Ed. (Ed.: Clarke, R. J.and Macare, R.).Elsevier, London and New York, p.83.

Ukers, W. H., 1935. *All About Coffee*, 2nd Ed. Tea and Coffee Trade Journal Co., New York.

United States Department of Agriculture (USDA), 2015. USDA Scientific Report of the Dietary Guidelines Advisory Committee.

Urgert, R., van der Weg, G., Kosmeijer-Schuil, T. G., van de Bovenkamp, P., Hovenier, R. and Katan, M. B., 1995. Levels of the cholesterol-elevating diterpenes cafestol and kahweol in various coffee brews.*Journal of Agricultural and Food Chemistry* 43(8), 2167–72.

Urgert, R. and Katan, M. B., 1997. The cholesterol-raising factor from coffee beans. *Annual Review of Nutrition* 17, 305–24.

USDA National Nutrient Database for Standard Reference (Release 28, released September 2015, slightly revised May 2016) United States Department of Agriculture, https://ndb.nal.usda.gov/ndb/.

USDA National Nutrient Database, 2017. United States Department of Agriculture, Agricultural Research Service, USDA Food Composition Databases, https://ndb.nal.usda.gov/ndb/, Accessed in February 13 2017.

van Boxtel, M. P. J. and Schmitt, J. A. J., 2004. Age related changes in the effects of caffeine on memory and cognitive performance. In: *Coffee, Tea, Chocolate and the Brain* (Ed.: Nehlig, A.).CRC Press, Boca Raton, Florida, pp. 85–97.

van Dijk A.E., Olthof, M. R., Meeuse, J. C., Seebus, E., Heine, R. J. and van Dam, R. M., 2009. Acute effects of decaffeinated coffee and the major coffee components chlorogenic acid and trigonelline on glucose tolerance.*Diabetes Care* 32, 1023–5.

Vitaglione, P., Fogliano, V. and Pellegrini, N., 2012. Coffee, colon function and colorectal cancer. *Food and Function* 3, 916–22.

Waite, M., 2015. Treatment for Alzheimer's disease: Has anything changed? *Australian Prescriber* 38, 60–3.

Wang W., Basinger A., Neese R. A., Shane B., Myong S. A., Christiansen M. and Hellerstein M. K. 2001. Am J Effect of nicotinic acid administration on hepatic very low density lipoprotein-triglyceride production. *Physiol Endocrinol Metab*. 280, 3, E540–7.

WHO/FAO (World Health Organization and Food and Agriculture Organization of the United Nations), 2002. Human vitamin and mineral requirements. Report of a Joint FAO/WHO Expert Consultation, Bangkok, Thailand.FAO, Rome. http://www.fao.org/docrep/004/y2809e/y2809e00.htm

WHO 2017, http://www.who.int/mediacentre/factsheets/fs310/en/accessed in January 2017.

Yamamoto, T., Yoshizawa, K., Kubo, S., Emoto, Y., Hara, K., Waters, B., Umehara, T., Murase, T. and Ikematsu, K., 2015. Autopsy report for a caffeine intoxication case and review of the current literature. *Journal of Toxicologic Pathology* 1, 33–6.

Yoshinari, O. and Igarashi, K., 2010. Anti-diabetic effect of trigonelline and nicotinic acid, on KK-Ay mice. *Current Medicinal Chemistry* 17(20), 2196–202.

Youngberg, M. R., Karpov, I. O., Begley, A., Pollock, B. G. and Buysse, D. J., 2011. Clinical and physiological correlates of caffeine and caffeine metabolites in primary insomnia. *Journal of Clinical Sleep Medicine* 7, 196–203.